Chains, Clearings, and Rebels: Solving a Family of Mode-Switching Subtraction Games

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Abstract

Mode-switching subtraction games are non-invariant subtraction games whose legal moves depend on the pile size n modulo a fixed m. In the diagonal family $D(m, r)$, a position $n \equiv r \pmod{m}$ permits the subtraction of any positive square at most n; every other position permits only m. We solve the family completely: outcomes and Grundy values under normal play, and outcomes under misère. For every $m \geq 3$ except $m = 4$ and every r, the P-positions are the $m - 1$ residue classes modulo $2m$ of the fallback chains' bottoms, together with the hard half of a finite set $\mathrm{Bad}(m, r)$ read off a single parameter $s_{\min}$ — the boundary position $n = 0$ when $r = 0$, and a single rebel $n = m+r \in \{13, 14, 15\}$ at $(m, r) \in \{(9, 4), (9, 5), (9, 6)\}$. Under misère play the chains invert and the exceptional positions are exactly the free half of the same set: one exception set, two conventions. The Grundy function on the square-mode class is, beyond the finite prefix, an explicit shift of the nim-sequence of Golomb's subtract-a-square game; consequently no member of the family is eventually periodic at the level of Grundy values, and each has bounded Grundy values if and only if Golomb's game does — a question that is open. At $m = 2$ and $m = 4$ the squares modulo $2m$ lie inside $\{0, 1, m\}$, the affected classes are confined, and they are Golomb's game in affine disguise, never eventually periodic, of density zero within their class. The mechanism is isolated as a Transfer Principle for arbitrary impartial games, and it runs backward: a host construction embeds any subtraction game whatever as a mode class of a mode-switching game, at outcome level and, with one added move, at Grundy level. So a classification theorem for mode-switching subtraction games with arbitrary mode sets would classify all subtraction games. The same architecture solves Foursquare, an earlier machine-found game with moves $\{3, 8, 9\}$ and a square mode at $n \equiv 1 \pmod{4}$, and yields an exact criterion for which move-set enlargements preserve a law. Every conjecture was machine-proposed from a census of roughly 3,000 games; twice, theory constructed afterwards rederived numbers the instrument had recorded before the theory existed. The growth rate of Golomb's P-set — his own open question since 1966 — remains open.

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1 Introduction

Take a pile of n counters and two rules. If $n \equiv 1 \pmod{3}$, the player to move may remove any positive square number of counters; from every other pile the only legal move is to remove exactly three. Players alternate, and whoever cannot move loses. That is the entire game, and it is completely solved: the player to move loses exactly when $n \equiv 0$ or $2 \pmod 6$. The law holds from $n = 0$ upward, with no preperiod and no exceptions.

Now change one number. Squares may be removed from any odd pile, and from an even pile a player removes exactly two. The even piles are tame: the player to move loses precisely when n is a multiple of four. The odd piles one more than a multiple of four are all winning positions. The rest are the interesting ones. An odd pile of the form $n = 4t + 3$ is, read by the index t, exactly Golomb's game of 1966 — the moves that can matter are subtract a positive square. Its losing positions never fall into any eventual period; they thin away to density zero; and how fast they thin has been open for years. Two rulesets, one line each. One is finished in a sentence, the other runs into an unsolved problem, and what separates them is a single number determined by the ruleset before a counter is moved. This paper computes that number and shows what follows once it is in hand.

A subtraction game is played on a pile of counters: a ruleset names the amounts a player may remove, the players alternate, and under normal play whoever cannot move loses. For a fixed finite ruleset the classical theory ends in one word, periodicity: the sequence of previous-player wins is eventually periodic, and the game solves itself by a finite computation [2]; see [3] for a recent survey. The live frontier is rulesets in which the set of legal moves varies with the position being moved from — position-dependent, heap-dependent, or non-invariant, move sets [1, 20, 21]. This paper studies a structured family at that frontier and solves it completely (outcomes and Grundy values under normal play, outcomes under misère), and then shows that the structure making the family solvable sits one construction away from universality: the same mode-switching format can host any subtraction game whatever.

The protagonist is the two-parameter family $D(m, r)$: from a position $n \equiv r \pmod m$ a player may subtract any positive square not exceeding n; from every other position the only move is to subtract m. The family is called diagonal because its members first surfaced along a diagonal of a census's parameter grid, one game per modulus. The main outcome law can be tasted before any machinery arrives: for every $m \geq 3$ except $m = 4$ and every r, the P-positions of $D(m, r)$ are exactly $m - 1$ residue classes modulo $2m$ — the bottoms of the game's fallback chains — together with a finite exception set read off a single parameter, with no thresholds and no unverified zones.

The classification rests on two lemmas and one number. The Chain Lemma decomposes every fallback class into an independent chain whose P-positions alternate upward from a stuck bottom — Grundy values 0 and 1, nothing else, uniformly across the board's foreign territory. The H-duality lemma matches amounts to landings: an amount is useful from the square-mode class exactly when its residue modulo $2m$ avoids 0 and m, and the useful residues split into a hard set H and its shift, serving the class's two residues in complementary pairs. The number is $s_{\min}(m, r)$, the least square whose residue lies in H. Its trichotomy is the paper's map: $s_{\min} \leq f + m$ for every healthy pair — all $m \geq 3$, $m \neq 4$ outside three exceptions — while $s_{\min} = 25$ at $(m, r) \in \{(9, 4), (9, 5), (9, 6)\}$ and $s_{\min} = \infty$ at $m \in \{2, 4\}$. The class members below $s_{\min}$ form the bad set $\mathrm{Bad}(m, r)$, of size one, two, or three, and the Diagonal Law is then a single sentence: the P-positions are the bottom classes together with the hard half of Bad. The boundary position $n = 0$ and the three rebels were never two kinds of exception; they are the hard members of Bad in their regimes.

Behind the outcome argument sits a mechanism worth isolating, and Section 6 states it for an arbitrary impartial game: if a set of positions has, off a finite bad prefix, external options realising exactly the Grundy values 0 and 1, then its Grundy values are the internal game's values shifted by 2, the periphery and the prefix invisible. Run forward on the diagonal family, the principle yields the complete Grundy solution (Section 8): on the square-mode class, beyond Bad, $G = 2+\Gamma$ along an explicit reindexing, where Γ is the nim-sequence of Golomb's subtract-a-square game and the offset is read from $\beta = |\text{Bad}|$ and the squarefree part of m. Two corollaries calibrate how far outside the classical theory the family lives: no member is eventually periodic at the level of Grundy values, and each has bounded values if and only if Γ is bounded — Golomb's Nim-dimension question (that is, whether its Grundy values fit inside a fixed finite set $\{0, 1, \ldots, 2^d - 1\}$), reproduced at every modulus.

The excluded moduli are not casualties but the first instance of a phenomenon. At $m = 2$ and $m = 4$ the squares modulo $2m$ collapse into $\{0, 1, m\}$, no move from the affected classes reaches a foreign P-position, and the classes turn inward: they are Golomb's game in affine disguise (Section 7), provably never eventually periodic, of density zero within their class. Section 9 shows the phenomenon is universal: for any subtraction game T, a mode-switching host with fallback m and mode set $\{1\} \cup 2mT$ reproduces T's P-set on an arithmetic progression, and adding the single move $m + 1$ upgrades the embedding to Grundy values, exact from the base point $f + m$. The consequence closes a door: a classification theorem for mode-switching subtraction games with arbitrary mode sets would classify all subtraction games, so no such theorem exists to be proved — one construction away from the solved family lies a class as wild as subtraction games themselves.

Misère play, which usually wrecks structure, here confirms it (Section 10). The chains invert, the two witnesses exchange jobs, and the misère P-positions are the top rungs together with the free half of the same set Bad — one exception set, two conventions, the convention selecting the half. At the degenerate moduli misère swaps which residue is confined, and the misère P-set of the class is the normal-play P-set translated by $\mp m$, the index set never moving.

Section 11 puts the machinery to work three times: $D(5, 2)$ solved in full as the worked example; Foursquare (the machine's first find, moves $\{3, 8, 9\}$ with a square mode at $n \equiv 1 \pmod 4$) solved with its mechanism exposed and Kadam's Extension established; and a criterion determining exactly which enlargements of a mode's move set preserve a law. Section 12 opens the instrument itself — proposer, engine, detectors, filter — and records the retrodictions in which theory built afterwards rederived numbers recorded before the theory existed.

The method deserves a plain statement on page one. Every conjecture in this paper was machine-proposed, drawn from a census of roughly three thousand games; the instrument that proposed them constructs no proofs, and Section 12 documents the pipeline, the verification regime, and the written protocol governing its use. The acknowledgments state in full the part that automated systems played in the mathematics and in the writing.

Section 2 fixes notation and the verification lemma; Sections 3 through 5 build chains, the $s_{\min}$ apparatus, and the Diagonal Law. Section 6 isolates the Transfer Principle; Section 7 proves confinement at $m = 2$ and 4; Sections 8 and 9 carry the principle forward to the Grundy solution and backward to the universal embedding. Section 10 settles misère play; Section 11 solves further games; Section 12 opens the instrument; and Section 13 closes what the census could not settle — and locates the question that remains, older than this paper.

2 Preliminaries

Definition 2.1 (Positions and outcomes)**.** A terminal position (no legal moves) is a P-position; a position is an N-position if and only if at least one legal move reaches a P-position; a position is a P-position if and only if every legal move reaches an N-position, the terminal case satisfying this clause vacuously.

Lemma 2.2 (Verification principle; see, e.g., [9])**.** *Let S be a set of positions of an impartial game under normal play. If (i) every terminal position lies in S, (ii) no move from a position in S leads to a position in S, and (iii) from every position not in S some move leads to a position in S, then S is exactly the set of* P*-positions.*

The general theory of finite subtraction games and their eventual periodicity is developed in [4].

Definition 2.3 (The diagonal games $D(m,r)$)**.** Let $m \geq 2$ and $0 \leq r \leq m-1$. $D(m,r)$ is played on the non-negative integers. From a position $n \equiv r \pmod{m}$ (*squares mode*) a move subtracts any square k^2 with $k \geq 1$ and $k^2 \leq n$. From any other position (*fallback mode*) the only move subtracts m, available when $n \geq m$. Play is normal.

Definition 2.4 (Squares modulo $2m$)**.** $\mathcal{Q}(2m)$ denotes the set of residues of squares modulo $2m$. The mirror tool, proved once in general: $(2m-k)^2 = 4m^2 - 4mk + k^2 \equiv k^2 \pmod{2m}$, since $4m^2 = 2m \cdot 2m$ and $4mk = 2k \cdot 2m$. So every table is $k = 1..m$ plus mirrors, plus the 0 from $k = 2m$. Two side facts fell out of the mirrors and are used below: for odd m, $m^2 - m = m(m-1)$ has the even factor $m-1$, so $m^2 \equiv m \pmod{2m}$; for even m, $m^2 = (m/2) \cdot 2m \equiv 0 \pmod{2m}$.

mod	$\mathcal{Q}(2m)$	complement
4	$\{0,1\}$	$\{2,3\}$
6	$\{0,1,3,4\}$	$\{2,5\}$
8	$\{0,1,4\}$	$\{2,3,5,6,7\}$
10	$\{0,1,4,5,6,9\}$	$\{2,3,7,8\}$
12	$\{0,1,4,9\}$	$\{2,3,5,6,7,8,10,11\}$
14	$\{0,1,2,4,7,8,9,11\}$	$\{3,5,6,10,12,13\}$
18	$\{0,1,4,7,9,10,13,16\}$	$\{2,3,5,6,8,11,12,14,15,17\}$
22	$\{0,1,3,4,5,9,11,12,14,15,16,20\}$	$\{2,6,7,8,10,13,17,18,19,21\}$

Table 1: The engines at every modulus: $\mathcal{Q}(2m)$ with complements.

The mod-4 and mod-8 rows are the diagnosis written out: $\mathcal{Q}(4) = \{0,1\}$ and $\mathcal{Q}(8) = \{0,1,4\}$ both sit entirely inside the triple $\{0,1,m\}$, so their complements swallow every value any hard set will ever contain (Section 4) — the degeneracy of $m = 2$ and $m = 4$ is visible in these two rows before any game logic starts. One eyebrow kept raised on purpose: the thinnest *odd* table is mod 18, thinned because 9 is itself a square and the rows $k = 3$ and $k = 9$ collide there — $m = 9$'s anomaly is foreshadowed in its own table.

3 The Chain Lemma

Lemma 3.1 (Chain Lemma)**.** *Fix integers $m \geq 2$ and $0 \leq r \leq m-1$, and let c be any residue with $0 \leq c \leq m-1$ and $c \neq r$. In $D(m,r)$, for every position $n \geq 0$ with $n \equiv c \pmod{m}$:*

$$n \text{ is a P-position} \iff n \equiv c \pmod{2m}.$$

Proof. Rungs. The positions of class c form the progression c, $c+m$, $c+2m$, $\ldots$; call $c+jm$ rung j. *Closure.* A rung is never $\equiv r \pmod{m}$, so its only allowed move is the fallback $-m$, legal exactly when $j \geq 1$ and landing on rung $j-1$; rung $0 = c < m$ has no move at all, and no move ever leaves the progression. *Independence.* The status of a position is defined recursively from its options alone — the definition never mentions which positions move in. Since Closure makes every option of a rung another rung, the chain computes its statuses as if the rest of the game had been deleted. *Alternation.* We claim rung j is P if and only if j is even. Base: rung 0 is terminal by Closure, hence P. Step: assume the claim for rung $j-1$; rung j's single option is rung $j-1$, and a single-option position is P if and only if its option is N — so for j even, rung $j-1$ is N by the assumption and rung j is P, while for j odd, rung $j-1$ is P and rung j is N. *Translation.* j is even exactly when $c + jm \equiv c \pmod{2m}$, so the P-positions of class c are precisely the $n \equiv c \pmod{2m}$. ☐

Corollary 3.2 (Chain Lemma, Grundy form)**.** *Let $c \neq r$ with $0 \leq c \leq m-1$. In $D(m,r)$, a position $n \equiv c \pmod{m}$ has Grundy value*

$$G(n) = 0 \textit{ if } n \equiv c \pmod{2m}, \quad \textit{and} \quad G(n) = 1 \textit{ if } n \equiv c+m \pmod{2m}.$$

In particular every position outside class r has Grundy value 0 or 1.

Proof. By the Closure and Independence steps of Lemma 3.1, class c is a path under the single move $-m$, and rung j has exactly one option, rung $j-1$, for $j \geq 1$, and none for $j = 0$. Hence $G(\text{rung } 0) = 0$ and $G(\text{rung } j) = \operatorname{mex}\{G(\text{rung } j-1)\}$ alternates between 0 and 1, so $G(\text{rung } j) = j \bmod 2$. Translating as in Lemma 3.1, j is even exactly when $n \equiv c \pmod{2m}$. ☐

We call a position of a foreign class a *bottom* if it is an even rung and a *top rung* if it is an odd rung; Corollary 3.2 says bottoms carry Grundy value 0 and top rungs carry Grundy value 1, uniformly across all foreign classes.

4 Escape: the parameter $s_{\min}$

Lemma 3.1 and Corollary 3.2 settle every class except r. This section builds the apparatus for class r itself: the two residues the class occupies modulo $2m$, the hard set of useful residues, and a single parameter, $s_{\min}$, from which the classification of Section 5, its exception sets, and the Grundy and misère laws of Sections 8 and 10 are all read.

Definition 4.1 (The two residues, the hard set, and $s_{\min}$)**.** Fix $m \geq 2$ and $0 \leq r \leq m-1$. Let f denote the representative of r modulo m lying in the interval $[1, m]$, so that $f = r$ when $r \geq 1$ and $f = m$ when $r = 0$, and set $h = f + m$. The class r splits modulo $2m$ into exactly the two residues f and h, which we call the *free* and *hard* residues; when $r = 0$ these are m and 0. Write

$$B = \{0, 1, \ldots, m-1\} \setminus \{r\}$$

for the set of *foreign bottoms* (the bottoms of the $m-1$ chains classified by Lemma 3.1), and define the *hard set*

$$H(m,r) = ([f+1,\ f+m] \setminus m\mathbb{Z}) \bmod 2m.$$

For $r \geq 1$ this is $\{r+1,\dots,m-1\} \cup \{m+1,\dots,m+r\}$, and for $r=0$ it is $\{m+1,\dots,2m-1\}$. Finally, let

$$s_{\min}(m,r) = \min\{\, k^2 : k \geq 1 \text{ and } k^2 \bmod 2m \in H(m,r)\,\},$$

with $s_{\min} = \infty$ when no such square exists.

The point of taking f in $[1,m]$ rather than in $[0,m-1]$ is that the free residue is not the smaller of the two representatives but the one from which the move -1 reaches the bottom layer: from residue x the move -1 lands on $x-1$, which lies in $[0,m-1]$ precisely when $x \in [1,m]$. At $r=0$ the representative 0 fails this and the representative m satisfies it, which is why the free residue there is m and the hard residue is 0. With this convention every statement below is uniform in r.

Lemma 4.2 (H-duality)**.** *With the notation of Definition 4.1:*

(i) The map $b \mapsto h-b$ is a bijection from B onto $H(m,r)$ modulo $2m$, and $b \mapsto f-b$ is a bijection from B onto $H(m,r)+m$.

(ii) $\mathbb{Z}/2m\mathbb{Z}$ is the disjoint union of $\{0,m\}$, $H(m,r)$, and $H(m,r)+m$.

(iii) Let n be a class-r position and a a positive integer with $a \leq n$. If $a \equiv 0$ or $m \pmod{2m}$, the landing $n-a$ again lies in class r. Otherwise, if $a \bmod 2m \in H(m,r)$, then a carries the hard residue onto a foreign bottom and the free residue onto a top rung; and if $a \bmod 2m \in H(m,r)+m$, it carries the free residue onto a foreign bottom and the hard residue onto a top rung.

(iv) $1 \in H(m,r)+m$ for every $m \geq 2$ and every r.

Proof. (i) The map is visibly injective modulo $2m$, so only its image is at issue. Take the integer representative $h^* = f+m \in [m+1, 2m]$. If b ranged over all of $[0,m-1]$, then h^*-b would range over the m consecutive integers $[f+1, f+m]$. Any m consecutive integers contain exactly one multiple of m, never none and never two. The single element of $[0,m-1]$ omitted from B is r itself, and its image is $h^*-r = f+m-r$, which equals m when $r \geq 1$ and $2m$ when $r=0$ — in both cases the unique multiple of m in the window. Hence the image of B is $[f+1, f+m] \setminus m\mathbb{Z}$, which is $H(m,r)$. Since $f \equiv h-m$, the second map is the first shifted by $-m \equiv m$, so its image is $H(m,r)+m$.

(ii) The windows $[f+1, f+m]$ and $[f+m+1, f+2m]$ are disjoint and their union consists of $2m$ consecutive integers, hence a complete residue system modulo $2m$. Each window contains exactly one multiple of m, and two multiples of m that are distinct modulo $2m$ are congruent to 0 and to m in some order. Deleting them leaves the images computed in (i).

(iii) If $a \equiv 0$ or $m \pmod{2m}$, then $a \equiv 0 \pmod{m}$ and the landing is again $\equiv r \pmod{m}$, so it lies in class r; note that $a \equiv m$ interchanges the free and hard residues while remaining internal to the class. Otherwise (ii) places $a \bmod 2m$ in H or in $H+m$. Suppose $a \equiv h-b$ with $b \in B$. From the hard residue the landing is $h-a \equiv b$, a foreign bottom; from the free residue it is $f-a \equiv h-m-a \equiv b+m$, the top rung of the same chain. The case $a \equiv f-b$ is the same computation with the roles of f and h exchanged. The Grundy values are those of Corollary 3.2.

(iv) Since $m \geq 2$, 1 is not a multiple of m. Moreover $1 \equiv 2m+1 \pmod{2m}$, and $2m+1$ lies in the window $[f+m+1, f+2m]$ precisely because $1 \leq f \leq m$: the left inequality requires

$f \le m$ and the right requires $f \ge 1$. Hence $1 \in H + m$ by (i). Concretely, $1 = f - (f-1)$ with $f - 1 \in [0, m-1]$ and $f - 1 \not\equiv f \equiv r \pmod m$, so $f - 1$ is a foreign bottom. $\square$

Remark 4.3. Three readings of Lemma 4.2, recorded once. First, the partition (ii) says an amount can leave class r exactly when its residue modulo $2m$ is neither 0 nor m: the exclusions formerly listed one by one are the two-element complement of $H \cup (H+m)$. Second, the hole in H sits at the residue $\equiv 0 \pmod m$ and is the image, under the bijection of (i), of the omitted bottom r — the bottom that class r would have if its members were in fallback mode; the gap in the hard set and the uselessness of the amounts $\equiv 0, m$ are one fact seen from two sides. Third, the two jobs of an amount in (iii) are complementary rather than independent because the free and hard residues differ by m: their landings under a common subtraction differ by m as well, and Corollary 3.2 assigns opposite values to residues differing by m. Nothing in the lemma uses that an amount is a square. Finally, since H and $H+m$ are disjoint and $1 \in H+m$ by (iv), the residue 1 does not lie in H; the square 1 therefore never certifies, and as no square lies strictly between 1 and 4, it follows that $s_{\min} \ge 4$ whenever $s_{\min}$ is finite. And since the residues 0 and m are outside H, $s_{\min}$ is never a multiple of m.

Corollary 4.4 (Palette)**.** *Suppose $s_{\min}(m,r) < \infty$, and let n be a class-r position with $n \ge s_{\min}$. Then among the options of n there is one of Grundy value 0 and one of Grundy value 1, both leaving class r.*

Proof. Both 1 and $s_{\min}$ are squares not exceeding n, hence legal moves from n, and neither is $\equiv 0$ or m modulo $2m$ — the first by Lemma 4.2(iv), the second by definition of $s_{\min}$. If n sits on the hard residue then, by Lemma 4.2(iii), the move $-s_{\min}$ lands on a foreign bottom and the move -1 on a top rung; if n sits on the free residue the two witnesses exchange roles. In either case the pair $\{1, s_{\min}\}$ realises both values 0 and 1. $\square$

One square, two jobs: the witness that lands the hard residue on a bottom (the escape that Section 5 converts into the classification) is at the same time the Grundy-0 half of the palette, which Section 8 converts into a transfer of Grundy values. The two developments consume the same hypothesis, the finiteness of $s_{\min}$, at the same threshold.

Lemma 4.5 (A square in the window)**.** *For every $m \ge 3$ and every $0 \le r \le m-1$, the interval $(f,\ f+m]$ contains a perfect square, with a single exception: at $(m,r) = (4,0)$ the interval $(4,8]$ contains none.*

Proof. Suppose $(f, f+m]$ contains no square, and let c be the largest integer with $c^2 \le f$. Then $(c+1)^2 > f + m$, and subtracting $c^2 \le f$ gives $2c + 1 > m$, so $c \ge m/2$ and $c^2 \ge m^2/4$. Hence $m^2/4 \le f$. If $r \ge 1$ then $f = r \le m - 1$, and $m^2/4 \le m - 1$ rearranges to $(m-2)^2 \le 0$, impossible for $m \ge 3$. If $r = 0$ then $f = m$, and $m^2/4 \le m$ forces $m \le 4$, leaving $m \in \{3, 4\}$: the window $(3, 6]$ contains 4, so no failure occurs at $m = 3$, while $(4, 8]$ contains no square, the named exception. $\square$

Proposition 4.6 (The $s_{\min}$ trichotomy)**.** *Let $m \ge 2$ and $0 \le r \le m-1$, and let $E = \{(9,4), (9,5), (9,6)\}$. Exactly one of the following holds:*

(a) (healthy) $s_{\min} \le f + m$ — the case for every $m \ge 3$ with $m \ne 4$ and $(m,r) \notin E$; moreover $s_{\min} > f$, so the certifying square lies in the window $(f, f+m]$;

(b) (exceptional) $f + m < s_{\min} < \infty$ — exactly the pairs $(m,r) \in E$, where $s_{\min} = 25$ against $f + m \in \{13, 14, 15\}$;

(c) (degenerate) $s_{\min} = \infty$ — exactly the moduli $m \in \{2, 4\}$, for every r.

Proof. The proof turns on one equivalence: $s_{\min} \le f+m$ if and only if the window $(f, f+m]$ contains a square other than m and $2m$. Indeed, a square $s \le f$ has residue s (as $s < 2m$), and every residue in H exceeds r when read in $[0, 2m)$ (its elements are $\{r+1, \dots, m-1\} \cup \{m+1, \dots, m+r\}$ for $r \ge 1$ and $\{m+1, \dots, 2m-1\}$ for $r = 0$), so $s \notin H$ and squares up to f never certify. A square $s \in (f, f+m]$ with $s < 2m$ has residue s, which lies in H exactly when s is not a multiple of m; the window contains exactly one multiple of m (any m consecutive integers do), namely m itself when $r \ge 1$ and $2m$ when $r = 0$, and if $s = 2m$ its residue is $0 \notin H$. The equivalence follows in both directions.

(c) At $m \in \{2, 4\}$, Table 1 gives $\mathcal{Q}(2m) \subseteq \{0, 1, m\}$. The residues 0 and m lie outside H by Lemma 4.2(ii), and 1 lies outside H by Remark 4.3; hence no square has residue in H and $s_{\min} = \infty$.

Now let $m \ge 3$, $m \ne 4$. By Lemma 4.5 the window contains a square, so by the equivalence, (a) can fail only if every square in the window is the multiple $\mu \in \{m, 2m\}$. If $\mu = 2m$, then $r = 0$ and the open interval $(m, 2m)$ is square-free; the argument of Lemma 4.5's $r = 0$ case (with c the largest integer satisfying $c^2 \le m$) forces $m \le 5$, and the direct checks at $m = 3$ and $m = 5$ — the squares $4 \in (3, 6)$ and $9 \in (5, 10)$ — rule both out. If $\mu = m$, then $m = b^2$ is a perfect square and the window's only square is m: no smaller square intrudes, $(b-1)^2 \le f$, and no larger one, $(b+1)^2 > f + m$, i.e. $f \le 2b$. The window $(b-1)^2 \le f \le 2b$ is nonempty exactly when $(b-1)^2 \le 2b$, i.e. $b \le 3$; $b = 2$ gives $m = 4$, excluded, and $b = 3$ gives $m = 9$ with $f = r \in \{4, 5, 6\}$ — the set E. So (a) holds for every $m \ge 3$, $m \ne 4$ with $(m, r) \notin E$.

Finally, on E: the squares below 25 have residues $1, 4, 9, 16$ modulo 18. For each $r \in \{4, 5, 6\}$, the set $H(9, r)$ contains $\{7, 10, 13\}$ and none of $1, 4, 16$ (which lie in $H + 9$) nor 9 (the residue m); while $25 \equiv 7 \in H$. Hence $s_{\min} = 25$ there, which exceeds $f + m = r + 9$ and is finite: regime (b), and nothing else. □

Regime (a) is the former Clearing Lemma in its final form: the escape square is affordable from the least hard member $f + m$ onward. Regime (b) delays that escape to the fourth class member, and regime (c) removes it — the three regimes are the paper's trichotomy of healthy, exceptional and degenerate, now read off a single parameter.

Definition 4.7 (The bad set)**.** For $0 \le r \le m-1$ write $n_t = r + tm$, $t \ge 0$, for the members of class r in increasing order; we call t the *class index.* Define

$$\mathrm{Bad}(m, r) = \{\, n \in \text{class } r : n < s_{\min}(m, r) \,\}, \qquad \beta = |\mathrm{Bad}(m, r)|,$$

and write Bad_f and Bad_h for the sets of free and hard members of Bad — those on the residues f and h modulo $2m$ respectively. When $s_{\min} = \infty$ (regime (c)), Bad is the whole of class r and $\beta = \infty$; the finite case is the one in use everywhere except Section 10.2.

Lemma 4.8 (No deep square below $s_{\min}$)**.** *Let $\sigma(m)$ denote the least perfect square divisible by $2m$. If $s_{\min}(m, r) < \infty$, then $\sigma(m) > s_{\min}(m, r)$. Consequently no member of* $\mathrm{Bad}(m, r)$ *has a legal square move divisible by $2m$, and every internal option of a member of* Bad *lies in* Bad *and on the opposite residue modulo $2m$.*

Proof. In regime (a), $s_{\min} \le f + m \le 2m$, and $s_{\min} \ne 2m$ since its residue lies in H while $2m$ has residue 0; so $s_{\min} < 2m \le \sigma(m)$. In regime (b), $s_{\min} = 25$ while $\sigma(9) = 36$, the squares $1, 4, 9, 16, 25$ being indivisible by 18. Regime (c) is excluded by hypothesis. For the consequence: a square divisible by $2m$ has size at least $\sigma(m) > s_{\min}$, hence exceeds every member of Bad and is illegal from all of them. An internal move — a square $\equiv 0 \pmod m$ — has residue 0 or m

modulo $2m$; the residue-0 case is exactly divisibility by $2m$, just excluded. So within Bad every internal move has residue m, and by Lemma 4.2(iii) it exchanges the two residues of the class while landing on a smaller class member, which lies in Bad since Bad is an initial segment of the class. $\square$

Proposition 4.9 (Bad: size, pattern, and values)**.** *Let $s_{\min}(m,r) < \infty$. Then:*

(i) $\mathrm{Bad}(m,r)$ consists of the class members n_t with $0 \le t < \beta$, where $\beta = \lceil (s_{\min} - r)/m \rceil \ge 1$; the members alternate between the two residues, beginning on the free residue when $r \ge 1$ and on the hard residue when $r = 0$. In the three finite regimes: $\beta = 1$ with $\mathrm{Bad} = \{r\} = \mathrm{Bad}_f$ for healthy pairs with $r \ge 1$; $\beta = 2$ with $\mathrm{Bad} = \{0, m\}$, $\mathrm{Bad}_h = \{0\}$, $\mathrm{Bad}_f = \{m\}$ for healthy pairs with $r = 0$; and $\beta = 3$ with $\mathrm{Bad} = \{r, m+r, 2m+r\}$, $\mathrm{Bad}_h = \{m+r\}$, $\mathrm{Bad}_f = \{r, 2m+r\}$ for $(m,r) \in E$.

(ii) Every member of Bad_f has Grundy value 1, and every member of Bad_h has Grundy value 0.

(iii) Every internal option of a member of Bad lies in Bad, on the opposite residue modulo $2m$.

Proof. (i) The class members increase with t, so those below $s_{\min}$ form an initial segment, of length $\beta = \lceil (s_{\min} - r)/m \rceil$, the count of $t \ge 0$ with $r + tm < s_{\min}$. That $\beta \ge 1$: for $r \ge 1$ the least member is r, and $s_{\min} > f = r$ by Proposition 4.6(a)–(b); for $r = 0$ the least member is $0 < 4 \le s_{\min}$ by Remark 4.3. Adding m flips the residue modulo $2m$ between f and h, and $n_0 = r$ is the free residue when $r \ge 1$ and the hard residue ($\equiv 0 \equiv h$) when $r = 0$, giving the alternation as stated. The regime values: for healthy $r \ge 1$, $r < s_{\min} \le r + m$ gives $\beta = 1$; for healthy $r = 0$, every square up to m has residue at most m, outside $H(m,0) \subseteq [m+1, 2m-1]$, so $s_{\min} > m$, while $s_{\min} < 2m$ as in Lemma 4.8 — hence $\beta = 2$, with $n_0 = 0$ hard and $n_1 = m$ free; for E, $s_{\min} = 25$ with $r \in \{4,5,6\}$ gives $2 < (25-r)/9 < 3$, so $\beta = 3$ with the free–hard–free pattern $n_0 = r$, $n_1 = m + r$, $n_2 = 2m + r$.

(iii) is Lemma 4.8's consequence, restated for citation.

(ii) Mutual induction along Bad in increasing order. Let $p \in \mathrm{Bad}$ and consider any legal square $k^2 \le p < s_{\min}$. By minimality of $s_{\min}$ its residue is not in H, so by the partition of Lemma 4.2(ii) it is 0, m, or lies in $H + m$. Residue 0 is impossible from Bad by Lemma 4.8. Residue m is internal: by (iii) it lands in Bad on the opposite residue. Residue in $H + m$ is external: by Lemma 4.2(iii) it lands a free position on a foreign bottom, of Grundy value 0 by Corollary 3.2, and a hard position on a top rung, of value 1 — and the move -1 is of this kind (Lemma 4.2(iv)), legal whenever $p \ge 1$.

If $p \in \mathrm{Bad}_f$, then $p \ge f \ge 1$, so the option set is nonempty; every external option is a bottom of value 0, and every internal option is a member of Bad_h, of value 0 by induction. Hence $G(p) = \mathrm{mex}\{0\} = 1$. If $p \in \mathrm{Bad}_h$ with $p \ge 1$, every external option is a top rung of value 1, and every internal option is a member of Bad_f, of value 1 by induction; hence $G(p) = \mathrm{mex}\{1\} = 0$. If $p = 0$ — occurring only at $r = 0$ — then p is terminal and $G(p) = 0$. $\square$

5 The Diagonal Theorem

The pieces are now on the table, and they assemble into the classification in a single verification.

Theorem 5.1 (Diagonal Law)**.** *Let $m \ge 3$ with $m \ne 4$, and let $0 \le r \le m-1$. Write $\mathrm{Bad}_h(m,r)$ for the set of hard members of $\mathrm{Bad}(m,r)$. The P-positions of $D(m,r)$ are exactly the elements* *of*

$$S = \{\, n \geq 0 : n \equiv b \pmod{2m} \text{ for some } b \in B \,\} \cup \mathrm{Bad}_h(m, r).$$

The law holds for every $n \geq 0$, with no preperiod and no unverified zones. The hypothesis on m enters exactly once, through the finiteness of $s_{\min}$ (Lemma 4.5 and Proposition 4.6).

Proof. We verify the three conditions of Lemma 2.2 for S.

(i) *Terminals.* A position outside class r is terminal exactly when it is the bottom rung of its chain, that is, one of the positions $b \in B$ themselves: these satisfy $b < m$, so the fallback is illegal, and they are not in squares mode. Each lies in S. A class-r position $n \geq 1$ has the legal square 1, so the only class-r terminal is $n = 0$, which is in class r exactly when $r = 0$; there it is the least member of the hard residue and, since $s_{\min} \geq 4$ (Remark 4.3), lies in $\mathrm{Bad}_h \subseteq S$. Every terminal position therefore lies in S.

(ii) *Closure.* From a foreign member of S the only move is the fallback $-m$, which by the Closure step of Lemma 3.1 stays in its own chain and lands on the top rung $\equiv b+m \pmod{2m}$, outside S. From a member p of Bad_h: every square legal at p is smaller than $s_{\min}$, hence has residue outside H by the minimality in Definition 4.1, hence, by the partition of Lemma 4.2(ii), has residue 0, m, or in $H + m$. Residue 0 is impossible: such a square is divisible by $2m$ and so of size at least $\sigma(m) > s_{\min} > p$, by Lemma 4.8. A square of residue m stays in class r and exchanges the two residues (Lemma 4.2(iii)), landing on a free position below p — a free member of Bad, outside S. A square of residue in $H + m$ carries the hard residue onto a top rung (Lemma 4.2(iii)), outside S. Hence no move from S lands in S.

(iii) *Escape.* A foreign top rung $n \equiv b+m \pmod{2m}$ satisfies $n \geq b+m \geq m$, so the fallback is legal and lands on the bottom residue b — an element of S. A free class-r position satisfies $n \geq f \geq 1$, and the move -1 lands on the foreign bottom $f - 1$ (Lemma 4.2(iv)), an element of S. A hard class-r position outside S is precisely one with $n \geq s_{\min}$; the move $-s_{\min}$ is legal for that very reason and, its residue lying in H, lands on a foreign bottom (Lemma 4.2(iii)) — an element of S. Every position outside S has a move into S.

By Lemma 2.2, S is exactly the set of P-positions of $D(m, r)$. □

Corollary 5.2 (Explicit form)**.** *By Proposition 4.9,* Bad *occupies the class indices $t = 0, \ldots, \beta - 1$, with the free residue at even t when $r \geq 1$ and at odd t when $r = 0$. Hence:*

for healthy (m, r) with $r \geq 1$: $\beta = 1$ and $\mathrm{Bad} = \{r\}$ is free, so $\mathrm{Bad}_h = \emptyset$ and the P-set is the $m - 1$ bottom classes alone;

for healthy (m, r) with $r = 0$: $\beta = 2$ and $\mathrm{Bad} = \{0, m\}$ with 0 hard and m free, so $\mathrm{Bad}_h = \{0\}$;

for $(m, r) \in E$: $\beta = 3$ and $\mathrm{Bad} = \{r, m + r, 2m + r\}$ with $m + r$ hard and the other two free, so $\mathrm{Bad}_h = \{m + r\}$, the position we call the rebel.

The statement of Theorem 5.1 therefore coincides with the earlier three-clause law: the boundary position and the rebels were never two kinds of exception, but the hard members of Bad *in their respective regimes.*

Remark 5.3 (Receipts). The closure argument, instantiated at the rebel 13 of $D(9, 4)$: the legal squares are $1, 4, 9$, of residues $1, 4, 9$ modulo 18; the first two lie in $H + 9 = \{1, \ldots, 4\} \cup \{14, \ldots, 17\}$ and land on the top rungs 12 and 9; the third has residue m and lands on the free Bad member 4. The escape argument at the first cleared member: $31 - 25 = 6$, a bottom. The boundary at $r = 0$: $n = 0$ is terminal, and the positions with a move onto it are the class-0 members $n = k^2$, all of which the theorem already declares N — via the escape $-s_{\min}$ for $n \geq s_{\min}$, and via membership of Bad's free part otherwise.

Remark 5.4 (The historical witness). *The odd-m witness identity, the original route.* For odd m, $(m-1)^2 = m^2 - 2m + 1 \equiv m^2 + 1 \equiv \mathbf{m+1} \pmod{2m}$, using $m^2 \equiv m$ from Definition 2.4; and $m+1$ belongs to every hard set, landing the hard residue exactly on bottom $r-1$ ($m-1$ when $r=0$). So the pair $(1^2, (m-1)^2)$ is a universal witness pair for all odd m at once — at the price of the threshold $n \geq (m-1)^2$. Proposition 4.6(a) supersedes it: a square in the window $(f, f+m]$ does the same job with legality free, for both parities, with no threshold. The identity remains the cleanest line in this development and the historical door to everything above.

6 The Transfer Principle

The classification of Section 5 is an outcome statement, and the argument that produced it — one witness landing on a P-position — is the standard one. This section isolates the mechanism in a form that computes Grundy values rather than outcomes, and does so for an arbitrary impartial game. Two consequences follow in later sections: the Grundy values of every solved member of the diagonal family are a shift of Golomb's nim-sequence, and an arbitrary subtraction game can be embedded as a mode class of a mode-switching host.

Lemma 6.1 (Absorption). *Let $O \subseteq \mathbb{Z}_{\geq 0}$ with $\{0,1\} \subseteq O$, and set $V = O \cap [2, \infty)$. Then*

$$\operatorname{mex}(O) = 2 + \operatorname{mex}(V - 2).$$

Proof. Since 0 and 1 belong to O, $\operatorname{mex}(O) = \min\{y \geq 2 : y \notin O\}$. Writing $y = 2 + x$ with $x \geq 0$, the condition $y \notin O$ is equivalent to $x \notin V - 2$, and the minimum transports accordingly. □

The lemma is elementary; its force lies in what the hypothesis $\{0,1\} \subseteq O$ buys. It buys two things at once. It raises the floor, so that a position meeting the hypothesis has value at least 2 and the values of such positions occupy a band disjoint from $\{0,1\}$. And it saturates the low band, so that any option carrying value 0 or 1 is a duplicate, and mex is blind to duplicates. Together these make a finite set of irregular positions of low value not merely distinguishable but invisible, and it is this invisibility that the next theorem converts into a clean recursion.

Theorem 6.2 (Transfer Principle). *Let a well-founded impartial game under normal play be given, with Grundy function G. Let C be a set of its positions; for $p \in C$, call an option of p* internal *if it lies in C and* external *otherwise. Let $\mathrm{Bad} \subseteq C$, and assume:*

(H1) (palette) every $p \in C \setminus \mathrm{Bad}$ has an external option of Grundy value 0 and an external option of Grundy value 1;

(H2) (ceiling) every external option of every $p \in C$ has Grundy value 0 or 1;

(H3) (prefix) every $p \in \mathrm{Bad}$ has $G(p) \leq 1$, and every internal option of a member of Bad lies in Bad.

Then $G(p) \geq 2$ for every $p \in C \setminus \mathrm{Bad}$, and the function $h = G - 2$ on $C \setminus \mathrm{Bad}$ satisfies

$$h(p) = \operatorname{mex}\{\, h(q) : q \text{ an internal option of } p \text{ with } q \notin \mathrm{Bad} \,\}.$$

That is, h is the Grundy function of the game played on $C \setminus \mathrm{Bad}$ using internal moves only: the external options and the whole of Bad are invisible to it.

Proof. The option relation is well-founded, so we may induct along it. Let $p \in C \setminus \mathrm{Bad}$ and assume the conclusion at every member of $C \setminus \mathrm{Bad}$ below p. Write O for the set of Grundy values of the options of p and $V = O \cap [2, \infty)$.

By (H1), p has external options of values 0 and 1, so $\{0,1\} \subseteq O$. By (H2), every external option of p contributes a value in $\{0,1\}$ and hence nothing to V. By the first clause of (H3), an internal option lying in Bad has value at most 1 and likewise contributes nothing to V. An internal option q with $q \notin \mathrm{Bad}$ lies in $C \setminus \mathrm{Bad}$ and is below p, so by the induction hypothesis $G(q) = 2 + h(q) \geq 2$, and it contributes exactly that value. Hence

$$V = \{\, 2 + h(q) : q \text{ an internal option of } p \text{ with } q \notin \mathrm{Bad} \,\},$$

so $V - 2 = \{h(q) : q \text{ internal}, q \notin \mathrm{Bad}\}$. Lemma 6.1 applies to O and gives

$$G(p) = \mathrm{mex}(O) = 2 + \mathrm{mex}(V - 2) = 2 + \mathrm{mex}\{h(q) : q \text{ internal}, q \notin \mathrm{Bad}\},$$

which is at least 2 and is the asserted recursion. □

Remark 6.3. The theorem is stated for an arbitrary impartial game rather than for $D(m, r)$ because it is used twice under opposite logical circumstances. In Section 8 the hypotheses are *verified* for a family already defined: C is the class r, Bad is the set of class-r positions below $s_{\min}$, and (H1), (H2), (H3) are supplied respectively by Corollary 4.4, Corollary 3.2, and Proposition 4.9. In Section 9 they are *arranged*: the host game is constructed so that its mode class satisfies them by design, and the conclusion is then read as an embedding of a prescribed subtraction game. A statement tied to the diagonal family would serve the first use and not the second, and the mechanism is in any case indifferent to squares, to the fallback amount, and to the modulus — nothing in the proof mentions them. The failure of (H1) is what distinguishes the degenerate moduli: at $m \in \{2, 4\}$ the palette available to the hard residue is the single value 1, and the conclusion weakens from a Grundy transfer to the outcome transfer of Section 7.

Remark 6.4. The two clauses of (H3) do different work, and only the first is consumed by the proof. The first prevents a member of Bad from masquerading as a good position: a single $p_0 \in \mathrm{Bad}$ with $G(p_0) = 2$ would inject a phantom value into V at every good position having p_0 as an option, and the identification of h with the internal Grundy function would fail there and everywhere above it. The second clause is not needed for Theorem 6.2 itself but for the step that follows every application of it — the identification of the internal game on $C \setminus \mathrm{Bad}$ with a translate of the internal game on C. Deleting a downward-closed set from a subtraction game and re-basing yields an isomorphic copy of the same game, shifted; deleting a set with holes yields a genuinely different game, since a surviving position may have lost some options and not others. We therefore record both clauses in the statement, so that the hypotheses are checked once and the conclusion is available in both forms. When $\mathrm{Bad} = \emptyset$ the hypothesis (H3) is vacuous, no shift arises, and h is the internal Grundy function on the whole of C — this is the situation of the embedding of Section 9 on its hard progression, where the host is built so that the palette is available from the base point onward.

Remark 6.5. Neither Lemma 6.1 nor Theorem 6.2 is special to the value 2. If $\{0, 1, \ldots, c-1\} \subseteq O$ and $V = O \cap [c, \infty)$, the same two lines give $\mathrm{mex}(O) = c + \mathrm{mex}(V - c)$; and if (H1) and (H2) are replaced by the requirement that the external options of a good position realise exactly the values $0, 1, \ldots, c-1$ and never exceed them, and (H3) by $G \leq c - 1$ on Bad, then the conclusion becomes $G \geq c$ on $C \setminus \mathrm{Bad}$ with $h = G - c$ satisfying the same internal recursion. We use only the case $c = 2$, in which the ceiling is the alternation of Corollary 3.2; the general form is recorded because a host whose fallback modes are not single-step chains will present a taller ceiling, and the transfer survives verbatim.

7 The degenerate moduli

Theorem 5.1 excludes exactly two moduli, and this section proves the exclusion is not a limitation of the method but a fact about squares: at $m = 2$ and $m = 4$ there is no rescue to find, and the affected classes turn inward.

Lemma 7.1 (Confinement Lemma)**.** *For $m \in \{2, 4\}$ and every r: $\mathcal{Q}(2m) \subseteq \{0, 1, m\}$, while $H(m, r)$ excludes 0, 1, and m (Definition 4.1). Hence no square move from a hard-residue position reaches a foreign bottom: every move stays inside class r or lands on an* N*-position of a foreign chain.*

Proof. For $m \in \{2, 4\}$ and every r: $\mathcal{Q}(2m) \subseteq \{0, 1, m\}$ (read directly off the mod-4 and mod-8 rows of Table 1), while every hard set $H(m, r)$ excludes 0, 1, and m (Definition 4.1). Hence from any hard-residue position, every legal square move either stays inside class r (a square $\equiv 0$ or m mod $2m$ is $\equiv 0$ mod m) or lands one below on the odd rung above a bottom (a square $\equiv 1$), an N-position — never on a P-position of a foreign chain. The argument is an only-options enumeration over $\mathcal{Q}(2m)$, the mirror image of the Independence step in the proof of Lemma 3.1 — there, fallback moves cannot leave a chain; here, square moves cannot leave the class except onto N. *Consequence:* the hard class's P/N structure is decided by a self-contained internal recursion, and its very first member already defies the chain law: 3 is P in $D(2, 1)$ and 5 is P in $D(4, 1)$, where escape, if it existed, would force N. Calibration: for $m = 2$ and $m = 4$ we prove confinement — no square door onto the foreign bottoms, so the class turns inward. We do **not** prove that these confined classes obey no eventual law of their own: "no period and no modulus up to 200 fits" is a reading from the search instrument, not a theorem, and their aperiodicity remains open. □

Remark 7.2 (Double condemnation)*.* The two degenerate moduli are exactly the two the parity-free engine cannot serve. At $m = 2$ the engine is silent: the always-a-square inequality degenerates ($(m - 2)^2 = 0$) and the clearing interval $(1, 3]$ is genuinely square-free. At $m = 4$ every class fails from the outside as well: no square has a residue in any hard set, which is regime (c) of Proposition 4.6, and $(4, 8]$ is the one such window at any $m \geq 3$. Outside view and inside view agree because both are reading the same two table rows: $\mathcal{Q}(2m) \subseteq \{0, 1, m\}$.

Theorem 7.3 (The confined classes are Golomb's game)**.** *Let $m \in \{2, 4\}$ and $0 \leq r \leq m - 1$, and let h^* be the least representative of the hard residue: $h^* = m + r$ if $r \geq 1$, and $h^* = 0$ if $r = 0$. Index the confined class of $D(m, r)$ by $n = 2mt + h^*$, $t \geq 0$, and let $\mathcal{G}$ denote the set of P-positions of Golomb's subtract-a-square game. Then for $m = 2$ the position $4t + h^*$ is P in $D(2, r)$ if and only if $t \in \mathcal{G}$, and for $m = 4$ the position $8t + h^*$ is P in $D(4, r)$ if and only if $\lfloor t/2 \rfloor \in \mathcal{G}$ — the confined class consisting of two interleaved copies of Golomb's game, one on each parity of t.*

Proof. (a) The three kinds of square move. Every $k \geq 1$ is odd, $\equiv 2 \pmod 4$, or a multiple of 4; we dispatch each kind from a hard-class position $n = 2mt + h^*$. If k is odd then $k^2 \equiv 1 \pmod 8$, hence $\equiv 1 \pmod{2m}$, and the landing $\equiv h^* - 1 \pmod{2m}$ is the odd rung directly above the foreign bottom $r - 1$ (bottom $m - 1$ when $r = 0$), an N-position by Lemma 3.1. If $k \equiv 2 \pmod 4$ then $k^2 \equiv 4 \pmod 8$: at $m = 2$ this is $\equiv 0 \pmod 4$ and the move stays in the class, while at $m = 4$ the landing $\equiv h^* - 4 \pmod 8$ is the free residue (r, or m when $r = 0$), every member of which is N because it subtracts 1^2 onto the foreign bottom directly beneath it — P by Lemma 3.1, legal from the smallest free member. If $4 \mid k$ then $2m \mid k^2$ and the move stays in the class. In the index t the internal moves read: at $m=2$, every even $k=2j$ subtracts $4j^2$, i.e. $t \mapsto t-j^2$; at $m=4$, every $k=4i$ subtracts $16i^2$, i.e. $t \mapsto t-2i^2$. Legality transfers exactly: since the amounts and $2mt$ are multiples of $2m$ while $0 \le h^* < 2m$, we have $4j^2 \le 4t+h^* \iff j^2 \le t$ and $16i^2 \le 8t+h^* \iff 2i^2 \le t$. So the internal option sets at index t are $\{\, t-j^2 : j^2 \le t \,\}$ at $m=2$ and $\{\, t-2i^2 : 2i^2 \le t \,\}$ at $m=4$ — verbatim the option sets of subtract-a-square and of subtract-$\{2i^2\}$. The three-way sort is genuinely three-way only at $m=4$: modulo 4 the values $k^2 \equiv 4$ and $k^2 \equiv 0$ collapse into one residue, so at $m=2$ parity alone decides — every even square stays internal — which is why $m=2$ hosts Golomb's game whole while $m=4$, whose middle class defects to the free residue, keeps only the doubled cousin subtract-$\{2i^2\}$. *(b) Terminals align.* At $m=2$ the index game has the single internal-terminal $t=0$; at $m=4$ it has $t=0$ and $t=1$ (the smallest internal move needs $2 \le t$) — matching the terminal sets of subtract-a-square and subtract-$\{2i^2\}$ exactly. For $r \ge 1$ these positions have real moves, but by (a) every one of them exits onto an N-position; for $r=0$ the alignment is literal: $h^*=0$, so $t=0$ is $n=0$, a moveless position — Golomb's own terminal in the flesh. *(c) The transfer.* The status of a position is determined by its options alone, and by (a) every exiting option is N; so a hard-class position is P iff all of its internal options are N, and N iff some internal option is P — exits can neither supply a P-witness nor block one. By induction on t, the map $t \mapsto \mathrm{status}(2mt+h^*)$ therefore satisfies exactly the recursion of the subtraction game with the option sets of (a). At $m=2$ that recursion is Golomb's, proving the first claim. *(d) The $m=4$ split.* In subtract-$\{2i^2\}$ every move is even, so the parity of t is invariant, and on each parity class the move $t \mapsto t-2i^2$ reads $u \mapsto u-i^2$ in the half-index $u=\lfloor t/2 \rfloor$ (legality $2i^2 \le t \iff i^2 \le u$ on both parities), with terminal $u=0$ on each side: two interleaved, independent copies of Golomb's game, so $8t+h^*$ is P iff $\lfloor t/2 \rfloor \in \mathcal{G}$. ☐

The anomaly ledger now reads as a codebook: $D(2,1)$'s recorded fingerprints $3, 11, 23$ are $4t+3$ over $t=0,2,5$ (the opening P-positions of Golomb's game, OEIS A030193), and $D(4,1)$'s $5, 13, 37$ are $8t+5$ over $t=0,1,4$, half-indices $u=0,0,2$: the same openers arriving through the two interleaved copies. The instrument had been filing Golomb's sequence under two aliases for the entire census — while re-deriving that very sequence as a calibration classic before every session.

Remark 7.4. Nothing here is special to squares: whenever a mode class of step $2m$ is confined, its internal game is the plain subtraction game on the index t with subtraction set $\{\, s/2m : s \in S,\ 2m \mid s \,\}$, where S is the mode's move set. This observation is developed in full in Section 9, where the internal game is prescribed rather than inherited.

Corollary 7.5 (No eventual period)**.** *For $m \in \{2,4\}$ and every r, the set of P-positions of $D(m,r)$ is not eventually periodic.*

Proof. Suppose some period p worked for all $n \ge N_1$. Membership would still repeat along any arithmetic progression, so restrict to the hard class $n = 2mt+h^*$: adding $\mathrm{lcm}(p,2m)$ to n preserves both the class and, beyond N_1, the P/N status, and in the index this says the set $\{t : 2mt+h^* \text{ is P}\}$ is eventually periodic with period $q = \mathrm{lcm}(p,2m)/2m$. At $m=2$ that set is $\mathcal{G}$ itself, the P-set of Golomb's game, by Theorem 7.3. At $m=4$ it is $\{t : \lfloor t/2 \rfloor \in \mathcal{G}\}$, and one parity suffices: restrict to even $t=2u$; membership of u in $\mathcal{G}$ repeats when t moves by $2q$, and $2q$ is a multiple of q, so $\mathcal{G}$ is eventually periodic with period q as well. Either way we are left holding: $\mathcal{G}$ is eventually periodic with period q beyond some N_0. Two facts kill this. The argument is folklore for any subtraction set containing a multiple of every integer; we include it to keep the paper self-contained. First, the P-positions of Golomb's game never run out —

one losing position per horizon. Suppose no P-position existed at or beyond some M. Take $v \geq M/2$ and $t = (v+1)^2 - 1$, which lies beyond M. Every legal square from t is at most v^2, because $(v+1)^2$ overshoots t by exactly 1; and $v^2 \leq t - M$, because $t - M = v^2 + 2v - M$ and $2v \geq M$. So every option of t lands at $t - k^2 \geq M$, all N by supposition, making t itself a P-position at or beyond M — contradicting the supposition. (The -1 in the construction does real work: at $t = (v+1)^2$ the square $(v+1)^2$ is legal, the move lands on 0, a P-position, and the position is honestly N.) Second, the kill. The first fact supplies a P-position t of Golomb's game with $t \geq N_0$. Periodicity walks membership up t's own progression: $t + q \in \mathcal{G}$, then $t + 2q \in \mathcal{G}$, and so on, every intermediate point lying at or above N_0, so every step is licensed. After qk^2 steps we reach $t + (qk)^2 \in \mathcal{G}$. But $(qk)^2$ is two things at once: a multiple of q, which is what let periodicity carry us there, and a perfect square, which makes the single move from $t + (qk)^2$ down to t legal. That move joins one P-position to another, which the definition of P forbids. No period exists. ∎

Corollary 7.6 (Sparse but never scarce)**.** *For $m \in \{2, 4\}$ and every r, the set of P-positions inside the confined class contains no two elements differing by a perfect square. Consequently it has asymptotic density zero, while its counting function up to n stays at least $\sqrt{n}/2 - O(1)$.*

Proof. The difference-free property is closure read twice: two P-positions of the class are both in squares mode, so if they differed by a square, the larger would have a legal move onto the smaller — a move from P to P, which is forbidden. Density zero is then the Furstenberg–Sárközy theorem [15, 16], which says exactly that a set of integers avoiding square differences has density zero. We claim no novelty at this bridge: for Golomb's game itself the observation and the connection are recorded by Eppstein [18], and through Theorem 7.3 our statement and his are one statement in two coordinate systems. The floor is Golomb's counting trick [2, 17], run on $\mathcal{G}$ and carried back through the reindexing. Every u in $[0, T]$ is either in $\mathcal{G}$ or wins by one square move onto some $g \in \mathcal{G}$ below it; a fixed g can serve at most $\sqrt{T}$ positions this way, one per square, plus itself. So $T + 1 \leq |\mathcal{G} \cap [0, T]| \cdot (\sqrt{T} + 1)$, giving $|\mathcal{G} \cap [0, T]| \geq \sqrt{T} - 1$. At $m = 2$ the confined positions up to n correspond to indices up to $T \approx n/4$, so at least $\sqrt{n}/2 - O(1)$ of them are P. At $m = 4$ there are two copies of $\mathcal{G}$, one per parity of t, living on disjoint sets of positions, so their floors add: each copy runs at $T \approx n/16$ and contributes $\sqrt{n}/4 - O(1)$, together $\sqrt{n}/2 - O(1)$ again. That both moduli land on the same constant is the small arithmetic accident $2/\sqrt{16} = 1/\sqrt{4}$. ∎

Corollary 7.7 (One density for the whole family)**.** *For every $m \geq 2$ and every r, the P-set of $D(m, r)$ has asymptotic density $(m-1)/2m$. For $m \geq 3$, $m \neq 4$ the correction to the periodic law is a finite set; for $m \in \{2, 4\}$ it is the confined class — infinite, of density zero, and slowly vanishing: by Corollary 7.6 it still holds at least $\sqrt{n}/2 - O(1)$ members up to n.*

Proof. For $m \geq 3$, $m \neq 4$, Theorem 5.1 lists the P-set as exactly $m - 1$ residue classes modulo $2m$ together with finitely many named positions, and $m - 1$ classes out of $2m$ carry density $(m-1)/2m$. For $m \in \{2, 4\}$ the same $m - 1$ classes arrive from the chains alone — Lemma 3.1 never needed the healthy hypothesis — the free residue contributes nothing, all of its members being N, and the confined class contributes density zero by Corollary 7.6. Written out at $m = 4$, where the accounting is least obvious: by Theorem 7.3 the confined P-set is $(16\mathcal{G} + h^*) \cup (16\mathcal{G} + h^* + 8)$, the even and odd parities of t. An affine map $u \mapsto 16u + b$ scales a counting function by the constant 16, so each image inherits density zero from $\mathcal{G}$, and a union of two density-zero sets is density zero — two, not infinitely many, is what makes that last step safe. ∎

8 The Grundy Transfer Theorem

Throughout this section $m \geq 3$, $m \neq 4$, and $0 \leq r \leq m-1$, so that $s_{\min}(m,r)$ is finite by Proposition 4.6 (the degenerate moduli, where $s_{\min} = \infty$, were treated in Section 7). We write Γ for the Grundy function of Golomb's subtract-a-square game, $\Gamma(u) = \operatorname{mex}\{\Gamma(u-j^2) : j \geq 1, j^2 \leq u\}$, so that $\Gamma(0) = 0$ and $\mathcal{G} = \Gamma^{-1}(0)$. The class index t of Definition 4.7, with $n_t = r + tm$, remains in force.

Definition 8.1. For $m = \prod p^{e_p}$ put $m_0 = \prod p^{\lceil e_p/2 \rceil}$ and $d = m_0^2/m$.

Proposition 8.2 (The internal game and the offsets)**.** *With the notation above:*

(i) (Internal moves.) A square k^2 is a move from class r into class r if and only if $m_0 \mid k$. Writing $k = m_0 j$, the amount is $m_0^2 j^2 = m \cdot dj^2$, so in the class index the internal moves read $t \mapsto t - dj^2$, legal exactly when $dj^2 \leq t$. The internal game is therefore the subtraction game with move set $\{dj^2 : j \geq 1\}$; it preserves t modulo d, and on the copy $\rho = t \bmod d$, in the half-index $u = \lfloor t/d \rfloor$, it reads $u \mapsto u - j^2$ — Golomb's game, d interleaved times.

(ii) (d is the squarefree part.) $d = \prod_{e_p \text{ odd}} p$, the squarefree part of m; in particular $d = 1$ if and only if m is a perfect square, and $d = m$ if and only if m is squarefree.

(iii) (Offsets.) By Proposition 4.9(i), Bad *occupies the class indices $0, \dots, \beta - 1$. Deleting* Bad *from the internal game and re-basing each copy, the copy $\rho \in \{0, \dots, d-1\}$ loses an initial run of exactly $a_\rho = \lceil (\beta - \rho)/d \rceil$ half-indices when $\rho < \beta$, and none when $\rho \geq \beta$.*

Proof. (i) Write $k = \prod p^{f_p}$. Then $m \mid k^2$ if and only if $e_p \leq 2f_p$ for every p, if and only if $f_p \geq \lceil e_p/2 \rceil$ for every p, if and only if $m_0 \mid k$. A square with $m \mid k^2$ satisfies $k^2 \equiv 0 \pmod{m}$, so subtracting it preserves the residue class r modulo m; conversely a square move that stays in class r must be $\equiv 0 \pmod{m}$. With $k = m_0 j$ the amount is $m_0^2 j^2 = (m_0^2/m) \cdot m j^2 = m \cdot dj^2$, and the landing is $r + (t - dj^2)m$. Legality: the move requires $m \cdot dj^2 \leq r + tm$, and since $0 \leq r < m$ and both $m \cdot dj^2$ and tm are multiples of m, this holds if and only if $dj^2 \leq t$. The move set $\{dj^2\}$ is invariant under translation, so t modulo d is preserved, and setting $u = \lfloor t/d \rfloor$ on a fixed copy turns $t \mapsto t - dj^2$ into $u \mapsto u - j^2$, with legality $dj^2 \leq t$ equivalent to $j^2 \leq u$ on that copy.

(ii) $m_0^2/m = \prod p^{2\lceil e_p/2 \rceil - e_p}$, and the exponent $2\lceil e_p/2 \rceil - e_p$ equals 0 when e_p is even and 1 when e_p is odd. Hence d is the product of the primes occurring to an odd power, which is squarefree, and $m = d \cdot (m/d)$ with m/d a perfect square. $d = 1$ says every exponent is even, i.e. m is a square; $d = m$ says every exponent is odd and the product of the odd-exponent primes is m, which forces every exponent to be 1.

(iii) The copy ρ consists of the class indices $t = \rho, \rho + d, \rho + 2d, \dots$, whose half-indices are $u = 0, 1, 2, \dots$. Such a t lies in Bad exactly when $\rho + ud < \beta$, that is $u < (\beta - \rho)/d$, that is $u \in \{0, 1, \dots, \lceil (\beta - \rho)/d \rceil - 1\}$. This is an initial run of the copy, of length $\lceil (\beta - \rho)/d \rceil$ when $\rho < \beta$ and empty when $\rho \geq \beta$. $\square$

The single copy at $d = 1$ absorbs the whole of Bad, which is why a perfect-square modulus shows a larger offset with no change of mechanism; the apparent anomaly at $m = 9$, $r = 0$ is this collapse and nothing more. The exceptional row is not a fit: $\beta = 3$ is forced by $s_{\min} = 25$, itself forced by the mod-18 square table in Proposition 4.6, and $a_0 = 3$ follows.

Corollary 8.3 (r enters only through β)**.** *For fixed m, the class-index Grundy sequence of $D(m,r)$ depends on r only through $\beta(m,r)$. Consequently all $m-1$ healthy games with $r \geq 1$ at a given modulus share one and the same class-index sequence, and the three exceptional games at $m = 9$ share one and the same class-index sequence.*

regime	d	β	Bad	a_0	a_1	a_ρ ($\rho \geq 2$)
healthy, $r \geq 1$	any	1	$\{r\}$	1	0	0
healthy, $r = 0$, m not a square	≥ 2	2	$\{0, m\}$	1	1	0
m a perfect square ($d = 1$), healthy	1	β	—	β	n/a	n/a
— of which $m = 9$, $r = 0$	1	2	$\{0, 9\}$	2	n/a	n/a
exceptional, $(m, r) \in E$	1	3	$\{r, m + r, 2m + r\}$	3	n/a	n/a

Table 2: The three regimes, with the offsets of Proposition 8.2(iii) evaluated. Here n/a marks the single-copy case $d = 1$, where no offset a_ρ with $\rho \geq 1$ exists.

Proof. The recursion of Theorem 8.4 below is determined by d, which depends only on m, and by the deleted initial segment, whose length is β; the external landings enter only through their Grundy values, which are 0 or 1 by Corollary 3.2 irrespective of which foreign chain they lie on. For the second claim, $\beta = 1$ for every healthy $r \geq 1$ by Proposition 4.9(i), and $\beta = 3$ for each member of E. $\square$

Theorem 8.4 (Grundy Transfer)**.** *Let $m \geq 3$, $m \neq 4$, and $0 \leq r \leq m - 1$; let d be as in Definition 8.1, β as in Definition 4.7, and a_ρ as in Proposition 8.2(iii). Then for every class-r position $n = r + tm$ with $n \geq s_{\min}(m, r)$,*

$$G(n) = 2 + \Gamma(\lfloor t/d \rfloor - a_\rho)\,, \qquad \rho = t \bmod d.$$

In particular $G(n) \geq 2$ there, and by Proposition 4.9(ii) the finitely many class-r positions below $s_{\min}$ carry the values 1 (free) and 0 (hard).

Proof. Apply Theorem 6.2 with C the class r and Bad $=$ Bad(m, r) of Definition 4.7. The induction of Theorem 6.2 requires the option relation on C to be well-founded, which it is: every move of $D(m, r)$ subtracts a positive amount, so options are positions of strictly smaller value in a set of non-negative integers.

(H1) holds off Bad: a class-r position with $n \geq s_{\min}$ has, by Corollary 4.4, external options of Grundy values 0 and 1, realised by the pair $\{1, s_{\min}\}$. (H2) holds: by Lemma 4.2(iii) an external option of a class-r position lies on a foreign bottom or a top rung, whose values are 0 and 1 by Corollary 3.2. (H3) holds by Proposition 4.9(ii)–(iii), whose two clauses are exactly the two clauses of the hypothesis.

Theorem 6.2 therefore gives $G \geq 2$ on $C \setminus$ Bad and $h = G - 2$ equal to the Grundy function of the internal game restricted to $C \setminus$ Bad. By Proposition 8.2(i) the internal game on the class index is subtract-$\{dj^2\}$, which splits into d copies indexed by $\rho = t \bmod d$, each reading as Golomb's game in the half-index $u = \lfloor t/d \rfloor$. By Proposition 4.9(i) the deleted set is an initial segment of the class index, hence by Proposition 8.2(iii) an initial run of length a_ρ within each copy; since the move set $\{j^2\}$ is translation-invariant, deleting an initial run and re-basing yields Golomb's game itself, shifted, so on copy ρ the function h at half-index u equals $\Gamma(u - a_\rho)$. Substituting $h = G - 2$ gives the statement. $\square$

Corollary 8.5 (Never eventually periodic)**.** *For every $m \geq 3$, $m \neq 4$, and every r, the Grundy sequence of $D(m, r)$ is not eventually periodic.*

Proof. Suppose it were, with period q beyond some N. Restrict to the class-r positions on a single copy ρ: adding $\mathrm{lcm}(q, dm)$ to n preserves the class and the copy (dm is the period of the copy structure in n, so any multiple of it does) and, beyond N, the Grundy value, so by Theorem 8.4 the sequence Γ is eventually periodic with period $\operatorname{lcm}(q, dm)/(dm)$. Its zero set $\mathcal{G}$ would then be eventually periodic, contradicting Corollary 7.5. ☐

Corollary 8.6 (Boundedness equivalence)**.** *For every $m \geq 3$, $m \neq 4$, and every r, the Grundy sequence of $D(m, r)$ is bounded if and only if Γ is bounded; and if bounded,* $\sup G = 2 + \sup \Gamma$.

Proof. Outside class r the values are 0 and 1 by Corollary 3.2; inside class r they are $2 + \Gamma(\cdot)$ beyond a finite set by Theorem 8.4, and the argument of Γ ranges over all sufficiently large integers on the copy $\rho = 0$, which is nonempty for every $d \geq 1$ since it contains $t = 0$. ☐

Whether Γ is bounded — equivalently, whether Golomb's game has bounded Nim-dimension — is open [6, 7]. Corollary 8.6 supplies an infinite family of mode-switching games each of which is equivalent to that question, and Corollary 8.5 places the whole solved family, and not only the two degenerate moduli of Section 7, outside the reach of the classical periodicity theory.

9 The Universal Embedding Theorem

Section 8 applied the Transfer Principle to a family already defined, verifying its hypotheses after the fact. This section reverses the order: given an arbitrary subtraction game, we build a mode-switching host whose hypotheses hold by construction, and read the conclusion as an embedding. The consequence is that the diagonal family's degenerate moduli were not an accident of squares. Any subtraction game whatever — periodic or aperiodic, of bounded or unbounded Nim-dimension — occurs as a mode class of a mode-switching game, at outcome level and, with one further move in the mode set, at Grundy level.

Definition 9.1 (The host)**.** Let $T \subseteq \mathbb{Z}^+$ be any nonempty set, and let $m \geq 2$ and $0 \leq r \leq m-1$. The *host* $H(T; m, r)$ is the mode-switching game on $\mathbb{Z}_{\geq 0}$ in which a position $n \equiv r \pmod{m}$ may have subtracted any amount of the mode set S, and every other position may have subtracted only m; play is normal. We consider two mode sets:

$$S_{\text{out}} = \{1\} \cup 2mT, \qquad S_{\text{Gr}} = \{1, m+1\} \cup 2mT,$$

where $2mT = \{2m\tau : \tau \in T\}$. Notation and terminology for the host are those of Definition 4.1 — f, h, B, bottoms, top rungs — which depend only on m and r and so apply verbatim; Lemma 3.1 and Corollary 3.2 likewise apply verbatim, since the foreign classes of the host are the same step-m chains.

Lemma 9.2 (The host's move geometry)**.** *In $H(T; m, r)$ with either mode set, and for every class-r position n:*

(i) Every amount $2m\tau \in 2mT$ is internal: the landing $n - 2m\tau$ lies in class r and on the same residue modulo $2m$.

(ii) The amount 1 lands the free residue on a foreign bottom and the hard residue on a top rung.

(iii) The amount $m+1$ lands the hard residue on a foreign bottom and the free residue on a top rung.

(iv) The class index t defined by $n = f^* + 2mt$, where f^* is the least member of the residue in question, transports the internal moves to $t \mapsto t - \tau$, legal exactly when $\tau \leq t$.

Proof. (i) $2m\tau \equiv 0 \pmod{2m}$. (ii) $1 \in H(m,r) + m$ by Lemma 4.2(iv), so Lemma 4.2(iii) applies. (iii) Since $1 \le f \le m$, the integer $m+1$ lies in $[f+1, f+m]$; and $m+1$ is not a multiple of m because $m \ge 2$. Hence $m+1 \in H(m,r)$ by Definition 4.1, and Lemma 4.2(iii) gives the two landings. (iv) The two residues f and h of class r each form an arithmetic progression of step $2m$, and an internal move subtracts $2m\tau$, so it moves t by τ; legality $2m\tau \le f^* + 2mt$ is equivalent to $\tau \le t$ since $0 \le f^* < 2m$. □

Note that the two residues of class r do not communicate: by (i) every internal move preserves the residue modulo $2m$, so the free half and the hard half of the class are separate boards. That is what makes the embedding clean; in the diagonal family the same disconnection occurs exactly when d is even, since internal moves preserve t modulo d (Proposition 8.2(i)) and the parity of t is the free–hard alternation (Proposition 4.9(i)).

Theorem 9.3 (Universal embedding, outcome level)**.** *Let $T \subseteq \mathbb{Z}^+$ be nonempty, $m \ge 2$, $0 \le r \le m-1$, and consider $H(T; m, r)$ with mode set S_{out}. Index the hard residue by $n = h^* + 2mt$, where h^* is its least member. Then the P-positions of the host are exactly*

the foreign bottoms, together with $\{h^* + 2mt : t \text{ is a P-position of the subtraction game } T\}$.

In particular the free residue contains no P-position, and the host's P-set is the set of foreign bottoms together with an affine copy of T's P-set.

Proof. The foreign classes are settled by Lemma 3.1. On the free residue, the move -1 is legal from the least free member onward (indeed always, since $f^* = f \ge 1$) and lands on a foreign bottom by Lemma 9.2(ii), a P-position; so every free member is N. On the hard residue, the external options are: the move -1, landing on a top rung, an N-position by Lemma 3.1; and nothing else, since S_{out} contains only 1 and internal amounts. Hence no external option of a hard member is a P-position, and no external option can be the P-witness of an N-position or an obstruction to a P-position. It follows by induction on n that a hard member is P if and only if all of its internal options are N, and N if and only if some internal option is P — which is the recursion of the subtraction game T under the index of Lemma 9.2(iv). The least hard member has index $t = 0$ with no internal option, matching T's terminal index 0. □

Corollary 9.4 (Section 7 recovered)**.** *Taking $T = \{j^2 : j \ge 1\}$ and $m = 2$ recovers Theorem 7.3: the confined class of $D(2, r)$ is Golomb's game in the index. More generally the confinement of Section 7 is the special case in which the host's internality is forced by a poverty of square residues rather than imposed by construction.*

Proof. At $m = 2$ the squares have residues in $\{0, 1\}$ modulo 4, so the mode set of $D(2, r)$ acts on class r exactly as S_{out} does with T the squares: the amounts $\equiv 0$ are internal, the amounts $\equiv 1$ land as in Lemma 9.2(ii). The index computation is that of Theorem 7.3. □

Theorem 9.5 (Universal embedding, Grundy level)**.** *Let $T \subseteq \mathbb{Z}^+$ be nonempty, $m \ge 2$, $0 \le r \le m-1$, and consider the host $H(T; m, r)$ with mode set $S_{\text{Gr}} = \{1, m+1\} \cup 2mT$. Write G_T for the Grundy function of the subtraction game T. Then:*

(i) (Hard residue: exact.) For every $t \ge 0$,

$$G(f + m + 2mt) = 2 + G_T(t).$$

When $r = 0$ the hard residue has one further member, $n = 0$, which is terminal and satisfies $G(0) = 0$.

(ii) (Free residue: one-step prefix.) $G(f) = 1$, and for every $t \geq 1$,

$$G(f + 2mt) = 2 + G_T(t - 1).$$

The embedding is thus exact on the progression $f + m + 2m\mathbb{Z}_{\geq 0}$ for every m and r, and exact on the free residue from its second member onward.

Proof. By Lemma 9.2(i) every internal move preserves the residue modulo $2m$, so the free and hard halves of class r are separate boards and Theorem 6.2 may be applied to each; the option relation is well-founded since every move subtracts a positive amount.

Free residue. Take $C = \{f + 2mt : t \geq 0\}$ and $\mathrm{Bad} = \{f\}$. At $n = f$ the amounts $m + 1$ and $2m\tau$ all exceed $f \leq m$, so the only option is $f - 1$, a foreign bottom of value 0 by Corollary 3.2; hence $G(f) = \mathrm{mex}\{0\} = 1 \leq 1$, and f has no internal option, so (H3) holds. For $t \geq 1$ we have $n = f + 2mt \geq 2m + 1 > m + 1$, so both witnesses are legal, and by Lemma 9.2(ii)–(iii) the amount 1 lands on a bottom (value 0) and $m + 1$ on a top rung (value 1): (H1). (H2) is Corollary 3.2. Theorem 6.2 gives $h = G - 2$ equal to the Grundy function of the internal game, the subtraction game T on the index t, by Lemma 9.2(iv), with the index 0 deleted; moves onto index 0 are excluded from the mex as moves into Bad. Deleting the initial segment $\{0\}$ and re-basing at $u = t - 1$ turns the legality $t - \tau \geq 1$ into $u - \tau \geq 0$, so $h(f + 2mt) = G_T(t - 1)$.

Hard residue, $r \geq 1$. Take $C = \{m + r + 2mt : t \geq 0\}$ and $\mathrm{Bad} = \emptyset$. The least member $m+r \geq m+1$ already affords both witnesses, so (H1) holds everywhere with the values exchanged ($m+1$ lands on a bottom, 1 on a top rung), and (H2) is as before. Theorem 6.2 with no deletion gives $G(m + r + 2mt) = 2 + G_T(t)$.

Hard residue, $r = 0$. Here the residue is 0 modulo $2m$ with members $2mt$, $t \geq 0$, and $n = 0$ is terminal: $G(0) = 0$. Take $\mathrm{Bad} = \{0\}$: the first clause of (H3) is $G(0) = 0$, and the second is vacuous, since 0 has no options at all. For $t \geq 1$, $n = 2mt \geq 2m \geq m + 1$, both witnesses are legal, and (H1), (H2) hold as in the previous case. Theorem 6.2 gives, after deleting the initial index 0 and re-basing at $t' = t - 1$, $G(2mt) = 2 + G_T(t - 1)$ for $t \geq 1$ — that is, writing $n = 2m + 2mt' = f + m + 2mt'$ with $f = m$, exactly the identity of (i). □

Remark 9.6. The two defects are the same defect: a class member too poor to afford its palette. On the free residue it is the least member $f \leq m$, which cannot afford $m+1$; at $r = 0$ on the hard residue it is $n = 0$, which can afford nothing at all and is the host's inheritance of the boundary position audited for the diagonal family in Section 5. Both are absorbed as one-element Bad sets, and both disappear under cosmetic changes we do not make: the theorem's content is the progression based at $f + m$, where the embedding is exact with no prefix for every m and r. This is the formal statement of the prefix-0 observation recorded by the search instrument, now with its base point made explicit.

Corollary 9.7 (What mode-switching games can do)**.** *For every nonempty $T \subseteq \mathbb{Z}^+$ there is a mode-switching subtraction game whose Grundy function, restricted to an arithmetic progression and shifted by 2, is that of T. Consequently there exist mode-switching subtraction games with aperiodic Grundy sequences of any type realised by a subtraction game, with unbounded Nim-dimension if any subtraction game has unbounded Nim-dimension, and with P-sets of any density realised by a subtraction game.*

Proof. Theorem 9.5 with the host $H(T; m, r)$. For the second sentence, a property of the Grundy sequence of T that is invariant under an affine reindexing and a shift by 2 transports to the host's class r; and outside class r the host's values are 0 and 1, which cannot restore periodicity to an aperiodic class, by the argument of Corollary 8.5. □

Remark 9.8. Taking $T = \{j^2 : j \geq 1\}$ in Theorem 9.5 gives a mode-switching game whose Grundy sequence is $2 + \Gamma$ on the hard residue, for **every** $m \geq 2$ — including $m = 2$ and $m = 4$, where the diagonal family itself achieves only the outcome-level embedding of Section 7 for want of the second palette witness. The move $m + 1$ is precisely what the squares fail to supply at those moduli, and supplying it by hand converts Theorem 7.3 into a Grundy statement. Taking instead T to be the move set of the aperiodic game of Larsson and Fox [7] produces a mode-switching subtraction game of Nim-dimension three — one more than T's own, since the host adds two to every value on the class — and taking T finite reproduces, inside a non-invariant host, the classical periodicity of [2] on one residue class while the rest of the board obeys the chain law.

Remark 9.9 (Scope). The construction uses only three properties of the host. The foreign classes are step-m chains, so that the ceiling of Theorem 6.2 is the alternation of Corollary 3.2; the mode set contains two amounts whose residues lie on opposite sides of the partition of Lemma 4.2(ii), so that the palette is available; and the remaining amounts are $\equiv 0 \pmod{2m}$, so that they are internal. A host whose fallback mode has a larger move set will fail the first of these and present a taller ceiling; by Remark 6.5 the Transfer Principle survives such a change with 2 replaced by the appropriate c, and the corresponding embedding theorem is the natural next question. It is not pursued here.

10 Misère play

Under the misère convention the player unable to move wins. A terminal position is therefore an N-position, and every other position is P if and only if all of its options are N. We write P^- and N^- for misère outcomes. No Grundy theory is available here. Misère outcomes of a disjunctive sum are not determined by the summands' outcomes [9], and the misère quotient of even a simple subtraction game may be large [19]. So this section classifies outcomes only. That is enough: the classification is complete, and the comparison with Theorem 5.1 is exact.

Lemma 10.1 (Chain Lemma, misère form). *Let $c \neq r$ with $0 \leq c \leq m - 1$. In $D(m, r)$ under misère play, a position $n \equiv c \pmod{m}$ is P^- if and only if $n \equiv c + m \pmod{2m}$. That is, the top rungs are P^- and the bottoms are N^- — the normal-play verdicts of Lemma 3.1, inverted.*

Proof. By the Closure and Independence steps of Lemma 3.1, class c is a path under the single move $-m$. Rung 0 is terminal, hence N^-. Rung $j \geq 1$ has the single option rung $j - 1$, so it is P^- exactly when rung $j - 1$ is N^-; by induction rung j is P^- exactly when j is odd, which is to say $n \equiv c + m \pmod{2m}$. □

The whole of the section follows from this one inversion, because it exchanges the values of the two witnesses of Corollary 4.4. In normal play the useful landing is a bottom, supplied to the hard residue by $s_{\min}$ and to the free residue by the always-legal 1; in misère the useful landing is a top rung, supplied to the *hard* residue by 1 and to the *free* residue by $s_{\min}$. The roles of the two witnesses are exchanged, and with them the roles of the two residues.

Lemma 10.2 (Misère Transfer Principle). *Let a well-founded impartial game be played under misère, let C be a set of its positions, and call an option of $p \in C$ internal if it lies in C and external otherwise. Suppose:*

(M1) no $p \in C$ is terminal;
(M2) every external option of every $p \in C$ is N^-.

Then for $p \in C$, p is P^- *if and only if every internal option of p is* N^- *— that is, the misère outcomes on C are the* normal-play *outcomes of the game played on C by internal moves alone.*

Proof. Let $p \in C$. By (M1) the terminal clause does not apply to p, so p is P^- exactly when all of its options are N^-; by (M2) the external ones are N^- unconditionally, so this holds exactly when all internal options are N^-. Reading "P^-" as the label of the internal game, the recursion is: a position of C is labelled P when every internal option is labelled N, and an internal-terminal position of C — one with no internal option — is labelled P vacuously. That is the normal-play recursion of the internal game. □

The last sentence is the point on which the section turns, and it is worth isolating: a position of C with no internal options is not terminal in the host, so the misère convention never reaches it, and it is labelled P by the vacuous clause as in normal play. The misère host contains the *normal-play* internal game, not its misère counterpart.

10.1 The healthy and exceptional games

Throughout this subsection $m \geq 3$, $m \neq 4$, so that $s_{\min} < \infty$; the notation f, h, $\mathrm{Bad}(m,r)$, Bad_f, Bad_h and β is that of Definitions 4.1 and 4.7.

Theorem 10.3 (Misère Diagonal Law)**.** *Let $m \geq 3$, $m \neq 4$, and $0 \leq r \leq m-1$. Under misère play the P-positions of $D(m,r)$ are exactly*

the top rungs of the foreign chains — the positions $n \equiv c + m \pmod{2m}$ for $c \in B$ — together with the free members of $\mathrm{Bad}(m,r)$*.*

In particular class r contributes $\lceil \beta/2 \rceil$ positions when $r \geq 1$ and $\lfloor \beta/2 \rfloor$ when $r = 0$; for a healthy game this is the single position $n = r$ when $r \geq 1$, and $n = m$ when $r = 0$, and for $(m,r) \in E$ it is the pair $n = r$ and $n = 2m + r$.

Proof. Foreign classes are settled by Lemma 10.1. For class r we take the four cases in turn.

Hard members. Let n lie on the hard residue. If $n = 0$ — possible only when $r = 0$ — then n is terminal, hence N^-. Otherwise $n \geq 1$, the amount 1 is legal, and by Lemma 4.2(iii) it lands on a top rung, which is P^- by Lemma 10.1; so n is N^-. Every hard member is therefore N^-, with no size condition and no exception.

Free members at or above $s_{\min}$. The amount $s_{\min}$ is legal and, having residue in H, carries the free residue onto a top rung, again P^-. So such n is N^-.

Free members of Bad. We induct on n within Bad. Let n be free with $n < s_{\min}$. By the definition of $s_{\min}$ no legal square of n has residue in H, so by Lemma 4.2(ii)–(iii) every legal square is either internal or of residue in $H + m$; the latter carries the free residue onto a bottom, which is N^- by Lemma 10.1. The internal options of n lie in class r below n, hence in Bad, and by Proposition 4.9(iii) they lie on the opposite residue; so they are hard members of Bad, N^- by the first case. Every option of n is therefore N^-, and n is P^-.

Hard members of Bad. Already covered by the first case; we note only that their N^- status was established without reference to Bad, so no circularity arises.

These cases exhaust class r, and the count follows from Proposition 4.9(i): Bad occupies the class indices $t = 0, \ldots, \beta - 1$, and the free indices are the even ones when $r \geq 1$ and the odd ones when $r = 0$. □

Corollary 10.4 (One exception set, two conventions)**.** *For $m \geq 3$, $m \neq 4$, the P-positions of $D(m,r)$ outside the periodic law are, in normal play, exactly the hard members of* $\mathrm{Bad}(m,r)$*,* *and in misère play exactly the free members of* Bad(m, r)*. The exception set is the same set under both conventions; the convention selects which half of it is exceptional.*

Proof. The misère half is Theorem 10.3. For normal play, Proposition 4.9(ii) assigns Grundy value 0 to the hard members of Bad and 1 to the free members, so the former are P-positions and the latter are not; and by Theorem 8.4 every class-r position at or above $s_{\min}$ has Grundy value at least 2. □

This is the strongest form of the observation that the rebels do not vanish under misère but move. In a healthy game with $r \geq 1$, Bad is the single free position $n = r$, so normal play has no exceptional P-position at all and misère has exactly one. At $r = 0$, Bad is $\{0, m\}$ with 0 hard and m free, so normal play has the boundary position $n = 0$ and misère has $n = m$. In the exceptional games Bad is $\{r, m+r, 2m+r\}$, of which $m + r$ is hard and the other two free: normal play keeps the rebel $m + r$, misère discards it and gains the pair r and $2m + r$. The receipts, at $(9, 4)$: the rebel 13 dissolves because its option 12 is a top rung and hence P^-, while 22 is P^- because its four options 21, 18, 13 and 6 are respectively a bottom, a bottom, a hard member of Bad, and a bottom — all N^-.

10.2 The degenerate moduli

At $m \in \{2, 4\}$ we have $s_{\min} = \infty$ and Bad is the whole of class r, so Theorem 10.3 does not apply and the analysis must be redone. It comes out as a mirror of Section 7, with the confined residue exchanged.

Proposition 10.5 (Misère confinement swaps residues)**.** *Let $m \in \{2, 4\}$ and $0 \leq r \leq m-1$. Let $f^* = f$ and let h^* be the least member of the hard residue, so that $h^* = f + m$ when $r \geq 1$ and $h^* = 0$ when $r = 0$. Let $\mathcal{G}$ denote the normal-play P-set of Golomb's subtract-a-square game, and set $I = \mathcal{G}$ when $m = 2$ and $I = \{u : \lfloor u/2 \rfloor \in \mathcal{G}\}$ when $m = 4$. Under misère play:*

(i) every hard member of class r is N^-;

(ii) the free members are confined, and $f^* + 2mu$ is P^- if and only if $u \in I$;

(iii) the misère and normal-play P-sets of class r are one index set read from two base points:

$$\mathrm{P}^- \cap (\textit{class } r) = \{f^* + 2mu : u \in I\}, \qquad \mathrm{P} \cap (\textit{class } r) = \{h^* + 2mu : u \in I\},$$

so that $\mathrm{P}^- \cap (\textit{class } r) = (\mathrm{P} \cap (\textit{class } r)) + (f^* - h^*)$, a translation by $-m$ when $r \geq 1$ and by $+m$ when $r = 0$.

Proof. (i) By Table 1 the squares have residues in $\{0, 1, m\}$ modulo $2m$ at both moduli. An odd square has residue 1, which lies in $H + m$ by Lemma 4.2(iv), so it carries the hard residue onto a top rung, P^- by Lemma 10.1; and an odd square is legal at every $n \geq 1$. The only hard member with $n = 0$ occurs at $r = 0$ and is terminal, hence N^-.

(ii) We verify (M1) and (M2) of Lemma 10.2 for the set C of free members. (M1): a free member satisfies $n \geq f \geq 1$ and so has the legal move 1; none is terminal. (M2): the options of a free member are of three kinds. A square of residue 1 lands on a bottom, N^- by Lemma 10.1. A square of residue m — occurring only at $m = 4$, where $4 \in \mathcal{Q}(8)$ — is internal but exchanges the two residues, so it lands on a hard member, N^- by (i). A square of residue 0 is internal and preserves the residue, so it is an internal option. Hence every external option of a member of C, and every option leaving C, is N^-, and Lemma 10.2 applies: the misère outcomes on C are the normal-play outcomes of the internal game on C.

That game is computed as in Theorem 7.3. At $m = 2$ a square divisible by 4 is $(2i)^2 = 4i^2$, so on the index u the internal moves read $u \mapsto u - i^2$, with legality $4i^2 \le f + 4u$ equivalent to $i^2 \le u$; this is subtract-a-square, whose normal-play P-set is $\mathcal{G}$. At $m = 4$ a square divisible by 8 is divisible by 16, hence of the form $(4i)^2 = 16i^2$, so the internal moves read $u \mapsto u - 2i^2$, with legality $16i^2 \le f + 8u$ equivalent to $2i^2 \le u$; this is subtract-$\{2i^2\}$, which preserves the parity of u and splits into two interleaved copies of subtract-a-square in the half-index $\lfloor u/2 \rfloor$, as in Theorem 7.3(d).

(iii) Theorem 7.3 states that the normal-play P-positions of class r are exactly $h^* + 2mt$ for $t \in I$. By (ii) the misère P^--positions are exactly $f^* + 2mu$ for $u \in I$. The index sets coincide because the two internal games are identical — subtract-a-square at $m = 2$, subtract-$\{2i^2\}$ at $m = 4$ — with index origin in each case at a position too small to afford any internal move, since $\max(f^*, h^*) < \sigma(m)$. And both are evaluated by the *normal-play* recursion: the hard side because play there is normal, the free side by Lemma 10.2, whose conclusion is precisely that the misère outcomes of the confined class are the normal-play outcomes of its internal game. Two sets built from one index set on two arithmetic progressions of the same step are translates of one another by the difference of base points, and $f^* - h^* = r - (m + r) = -m$ for $r \ge 1$, while $f^* - h^* = m - 0 = +m$ for $r = 0$. ☐

The sign is not a case split in the mechanism but in the geography: for $r \ge 1$ the free residue sits below the hard one, at $r = 0$ above it, and the index set never moves.

Corollary 10.6. *For $m \in \{2, 4\}$ and every r, the misère P-set of $D(m, r)$ is not eventually periodic; the confined free class contains no two elements differing by a perfect square, hence has asymptotic density zero, while retaining at least $\sqrt{n}/2 - O(1)$ members up to n; and the misère P-set has asymptotic density $(m - 1)/2m$.*

Proof. Each statement is the corresponding statement of Corollaries 7.5, 7.6 and 7.7 read through the reindexing of Proposition 10.5(ii), which differs from that of Theorem 7.3 only by the translation $f^* - h^* = \mp m$. The square-difference-free property transfers because two P^- positions of the confined class are both in squares mode and their difference, were it a square, would be a legal move from one to the other — a move between two P-positions of the misère internal game, which the normal-play recursion of Lemma 10.2 forbids. ☐

Remark 10.7. The misère picture is therefore complete for the whole family and follows the normal-play picture clause for clause: chains classified with values inverted, class r settled by a single parameter, exceptional positions confined to Bad, and the two degenerate moduli confined and equivalent to Golomb's game. What is not claimed is any statement about sums, for the reason given at the head of the section; the misère quotient of $D(m, r)$ is not computed here and appears to be a genuine question.

11 Further solved games

Example 11.1 ($D(5, 2)$ solved)**.** The P-positions of $D(5, 2)$ are exactly the $n \equiv 0, 1, 3, 4 \pmod{10}$.

Proof. Let $S = \{\, n \ge 0 : n \equiv 0, 1, 3, 4 \pmod{10} \,\}$; we verify the three conditions of Lemma 2.2. *(i) Terminals.* A squares-mode position — $n \equiv 2 \pmod 5$ — always has the legal move 1^2, since the smallest such position is 2; a fallback position has its only move -5 exactly when $n \ge 5$. The terminal set is therefore the set of fallback positions below 5, namely $\{0,1,3,4\}$ — the four chain bottoms — and all of them lie in S. *(ii) Closure.* Every element of S is $\equiv 0,1,3,4 \pmod{10}$, hence $\equiv 0,1,3,4 \pmod 5$: no candidate lies in class 2, so a candidate's only move is the fallback -5, whose landings occupy the residues $5,6,8,9 \pmod{10}$ — all outside S. (Legality plays no role here: a move that does not exist cannot violate closure.) No move from S returns to S. *(iii) Escape.* The complement of S consists of the residues $\{5,6,8,9\} \pmod{10}$ — the odd rungs of the four chains — together with $\{2,7\} \pmod{10}$, which is class 2. The odd rungs escape by the fallback -5, landing on $\{0,1,3,4\} \subseteq S$, legal because the smallest member of each class is at least 5. Class 2 escapes by its two keystones: from $n \equiv 2 \pmod{10}$ subtract 1^2, landing on residue 1, legal from the smallest member 2; from $n \equiv 7 \pmod{10}$ subtract 2^2, landing on residue 3, legal from the smallest member 7. Every position outside S has a move into S — and the verification consumes only the two instances $1^2 \equiv 1$ and $2^2 \equiv 4 \pmod{10}$; no general fact about squares is invoked. By Lemma 2.2, S is exactly the set of P-positions of $D(5,2)$: the law holds for every $n \geq 0$, with no preperiod — and the P-set consists of the bottom classes alone, for the reason recorded in Remark 11.3. $\square$

Remark 11.2 (Verification against the workbook). Prediction first, then the rows: residue-2 positions $(2,12,22)$ should be rescued by any legal square $\equiv 1$ or $9 \pmod{10}$, residue-7 positions $(7,17)$ by any square $\equiv 4$ or $6 \pmod{10}$, and every other square should be wasted.

- $n=2$: only move $2-1=1$ (P). Predicted rescue 1^2 present ✓, nothing else in the row to waste.
- $n=7$: $7-1=6$ (N), $7-4=3$ (P). Rescue 2^2 present ✓; the 1^2 is wasted as the congruence says ($1 \notin \{4,6\}$ mod 10, landing $\equiv 6$).
- $n=12$: $12-1=11$ (P), $12-4=8$ (N), $12-9=3$ (P). Primary rescue 1^2 ✓, and the spare $9=3^2$ makes its first appearance in the first row big enough to afford it.
- $n=17$: $17-1=16$ (N), $17-4=13$ (P), $17-9=8$ (N), $17-16=1$ (P). Primary 2^2 ✓ and spare 4^2 ✓.
- $n=22$: $22-1=21$ (P), $22-4=18$ (N), $22-9=13$ (P), $22-16=6$ (N). Primary 1^2 ✓, spare 3^2 ✓.

The check came back stronger than required: not only does the predicted rescue sit in every row, every wasted square is wasted for the predicted reason — its residue misses the survivor set, and every wasted landing is $\equiv 6$ or $8 \pmod{10}$, both N-residues. The congruence machinery reproduces the workbook letter-for-letter, and the 200,000 mechanical sweep stands as the bulk consistency evidence behind it.

Remark 11.3 (The $D(5,2)$ coincidence, resolved). For every game with finite $s_{\min}$ the P-set is, by Theorem 5.1, the union of the bottom classes and Bad_h. When Bad_h is empty — every healthy game with $r \geq 1$ — the P-set is the bottom classes alone, which is the coincidence observed above for $D(5,2)$; at $r=0$ and on E the set Bad_h contributes its single position; and at the degenerate moduli, where $s_{\min}=\infty$, the class-r contribution is the infinite confined set of Section 7.

Theorem 11.4 (Foursquare)**.** *Consider the game on $n \geq 0$: if $n \equiv 1 \pmod 4$, subtract any positive square $\leq n$; otherwise subtract $s \in \{3,8,9\}$, $s \leq n$; normal play. Then n is a* P*-position if and only if $n \equiv 0,2,4,$ or $6 \pmod{16}$, for all $n \geq 0$ with no preperiod.*

Proof of Theorem 11.4. Let $S = \{\, n \geq 0 : n \equiv 0,2,4,6 \pmod{16} \,\}$; since S and its complement are unions of residue classes modulo 16, every position lies in exactly one of the two. We verify the three conditions of Lemma 2.2 for S. *(i) Terminals.* The positions 0 and 2 are standard-mode (both $\not\equiv 1 \bmod 4$) and every allowed move $3, 8, 9$ exceeds them, so both are terminal. Conversely, every standard-mode $n \geq 3$ has the legal move 3, and every square-mode n has the legal move 1^2, since square mode requires $n \equiv 1 \pmod 4$ and hence $n \geq 1$. The terminal set is therefore exactly $\{0, 2\}$, and both members lie in S. *(ii) Closure.* Every element of S is even, hence $\not\equiv 1 \pmod 4$, hence standard-mode, so its moves are among $3, 8, 9$. The moves 3 and 9 are odd, and even minus odd is odd, so their landings occupy odd residues modulo 16 — all outside the all-even S. The move 8 carries the residues $\{0, 2, 4, 6\}$ onto $\{8, 10, 12, 14\}$, also outside S. (Legality plays no role here: a move that does not exist cannot violate closure.) Hence no move from S lands in S. *(iii) Escape.* The complement of S consists of the even residues $\{8, 10, 12, 14\}$ together with the odd residues, and the odd residues split by mode: $\{3, 7, 11, 15\}$ ($\equiv 3 \bmod 4$) are standard, $\{1, 5, 9, 13\}$ ($\equiv 1 \bmod 4$) are square. The even non-candidates escape by subtracting 8, landing on $\{0, 2, 4, 6\} \subseteq S$, legal because the smallest member of each class is at least 8. The standard-mode odds escape by $3 \to 0$ (residue 3), $3 \to 4$ (residue 7), $9 \to 2$ (residue 11), and $9 \to 6$ (residue 15), legal at the smallest members $3, 7, 11, 15$ respectively and a fortiori above. The square-mode odds escape by the squares 1^2 and 3^2: subtracting 1 carries the residues $\{1, 5\}$ to $\{0, 4\}$, and subtracting 9 carries $\{9, 13\}$ to $\{0, 4\}$, legal since $1 \leq n$ whenever square mode applies and $9 \leq n$ on classes whose smallest members are 9 and 13. Every position outside S therefore has a move into S — and the verification consumes only the two instances $1^2 \equiv 1$ and $3^2 \equiv 9 \pmod{16}$; no general fact about squares is invoked. By Lemma 2.2, S is exactly the set of P-positions of Foursquare: the law $n \equiv 0, 2, 4, 6 \pmod{16}$ holds for every $n \geq 0$, with no preperiod. $\square$

Remark 11.5. Every odd square is congruent to 1 or 9 modulo 16: writing $(2m+1)^2 = 4m(m+1) + 1$, the product $m(m+1)$ is even, so $4m(m+1) \equiv 0$ or $8 \pmod{16}$. The proof above needs only the two instances $1^2 \equiv 1$ and $3^2 \equiv 9$; the general statement is recorded because the mechanism remark below leans on it. The full Grundy-value sequence of Foursquare is OEIS A396951.

The question of which moves may be added to a subtraction game without changing it is an old one: Austin answered it in 1976 for purely periodic subtraction games. For each move s already present, the move $p - s$ may be adjoined, since the reply s reverses it out and the period p absorbs the pair. And Ho, in 2015, gave the idea its modern name, expansion sets [13, 14]. Proposition 11.6 and Remark 11.10 answer the same question for a mode-switching game, at the level of P-positions: the additions Foursquare tolerates are exactly the odd moves and the governor's clones $\equiv 8 \pmod{16}$, and the parity-and-residue argument that certifies them does for mode-switching games the job that reversibility did classically.

Proposition 11.6 (Kadam's Extension). *Let F' be Foursquare with its standard-mode move set enlarged from $\{3, 8, 9\}$ to every $s \equiv 3, 8,$ or $9 \pmod{16}$ with $s \leq n$, square mode unchanged. Then the P-positions of F' are exactly $n \equiv 0, 2, 4, 6 \pmod{16}$ — identical to Foursquare's. (Conjectured by Balaji Kadam, personal communication, July 2026.)*

Proof. Adding moves can only destroy terminality, never create it, so the terminals of F' lie among Foursquare's terminals $\{0, 2\}$, and both survive because the smallest member of the extended set is still $3 > 2$ — the terminal set is $\{0, 2\}$, inside the candidate $\{0, 2, 4, 6\} \pmod{16}$. From a candidate position — even, hence standard mode — every legal move of F' subtracts some $s \equiv 3, 8,$ or $9 \pmod{16}$. And the closure argument of Theorem 11.4 consumed exactly two facts about s, its parity and its residue modulo 16, never its actual size, so it transfers verbatim to every member of the extended family. The $s \equiv 3$ and $s \equiv 9$ moves are odd and land on odd residues, none of which lie in the all-even candidate, while the $s \equiv 8$ moves land on $\{8, 10, 12, 14\} \pmod{16}$, also outside — no move from the candidate returns to it. In the other direction, the extended move set contains the original one and square mode is untouched, so every escape witness of the original proof (the standard moves and the squares 1 and 9) remains a legal move of F' with the same landing, and a move set that only grows cannot lose an escape. Every position outside the candidate still has its move into it. The candidate therefore meets all three conditions of Lemma 2.2 in F' (terminals inside, closed against return, reachable from everywhere outside), so it is the P-set of F', and it coincides with Foursquare's. □

Proposition 11.7 (Which enlargements are harmless)**.** *Let a mode-switching game obey a law with no preperiod, and enlarge the move set of one mode by an amount a. The law survives if and only if no P-position has another P-position exactly a below it, with both positions in the enlarged mode.*

Proof. Adding moves can only affect closure, never escape: every old witness remains legal with the same landing, so condition (iii) of Lemma 2.2 survives untouched, and condition (i) can only shrink the terminal set, which the hypothesis of no preperiod already places inside the candidate. So the law survives precisely when closure does. Forward: if no P-position of the enlarged mode sits a above another P-position, then no new option leaves a P for a P, closure holds, and Lemma 2.2 returns the same P-set. Converse: let S be the P-set of the original game, suppose some position of the enlarged mode lies in S with its option a below also in S, and let n_0 be the smallest such position. We claim every position below n_0 keeps its original status in the enlarged game, by strong induction. A position $n < n_0$ with $n \in S$ has all of its old options outside S, hence N-positions by the induction hypothesis; its possible new option $n - a$ (present only when n is in the enlarged mode) cannot lie in S, since that would contradict the minimality of n_0; so every option of n is an N-position and n remains a P-position. A position $n < n_0$ outside S keeps its old witness into S, still a P-position by the induction hypothesis, so n remains N. At n_0 itself the new option $n_0 - a$ lies below n_0 and in S, hence is a P-position of the enlarged game, so n_0 is an N-position: the law fails, and it fails first at n_0. □

Remark 11.8 (The scoping is not decorative)*.* The criterion asks about P-positions of the enlarged mode, not about all P-positions. Enlarging $D(5, 2)$'s squares-mode set by the amount 2 leaves its law untouched, and provably so, no longer by sweep. By Corollary 5.2 the class 2 of $D(5, 2)$ contains no P-position at all, so the criterion's condition holds vacuously and Proposition 11.7 returns the law unchanged. This holds even though 2 is a difference of two P-positions of the game ($2 = 2 - 0$ modulo 10). An unscoped criterion predicts a break here and is simply wrong.

Corollary 11.9 (Kadam's Extension, revisited)**.** *In Foursquare every P-position is standard-mode, so the scoping is invisible and the criterion reduces to a residue condition: an added amount is harmless exactly when it is not congruent modulo* 16 *to a difference of two elements of* $\{0, 2, 4, 6\}$*. Those differences are* $\{0, 2, 4, 6, 10, 12, 14\}$*, so the harmless amounts are exactly the odd ones and those* $\equiv 8 \pmod{16}$ *— the governor's clones. The enlargement of Proposition 11.6 to the full residue classes of* 3*,* 9 *and* 8 *lies inside this set, which is why the P-set cannot move. A sweep over every added amount from* 1 *to* 32 *matches the criterion's prediction, and the breaks land where the converse says they must, at* $n = 2, 4, 6$ *and* 16 *for the amounts* $2, 4, 6$ *and* 16*. One outlier is instructive: adding* 10 *breaks first at* $n = 16$*, not* $n = 10$*, because* $10 = 16 - 6$

needs the larger P-position to exist before the pair can fire — the converse promises the smallest violating position, not the smallest position congruent to anything.

Remark 11.10 (The mechanism: why 16, and why $\{3, 8, 9\}$). Foursquare's board splits three ways, because square mode lives only at $n \equiv 1 \pmod 4$: the evens, the standard-mode odds $\equiv 3, 7, 11, 15 \pmod{16}$, and the square-mode odds $\equiv 1, 5, 9, 13$. Within the evens, sort the standard moves by parity and the game simplifies at a stroke: 3 and 9 land among the odds, never on the all-even P-set, so the single even move 8 governs the even world alone. That world under one jump of 8 is precisely the shape of the Chain Lemma (Lemma 3.1) — four chains by residue modulo 8, with bottoms $0, 2, 4, 6$ too poor to afford the jump, and a jump of size m minting a law of modulus $2m$. So 8 mints 16, with P at the even rungs: $n \equiv 0, 2, 4, 6 \pmod{16}$, Theorem 11.4, its four residues revealed as chain bottoms; and running the law modulo 8 instead kills closure on the spot, $8 \to 0$ being a move from the candidate into itself. And 3 and 9 are not decoys. Delete them and the law breaks at once, first at $n = 3$. For on the standard-mode odds, where squares are illegal, they are the sole rescuers: residue 3's only witness is 3 (to bottom 0), residue 7's is 3 (to 4), residue 11's is 9 (to 2), residue 15's is 9 (to 6). While the squares police only their legal home, the value 1 carries residues $\{1, 5\}$ to bottoms 0 and 4, and the value 9 carries $\{9, 13\}$ to the same two bottoms. Here is the number-theoretic heart at its true address: odd squares modulo 16 take exactly two values, 1 and 9 (Remark 11.5), and the second is load-bearing precisely at residues 9 and 13 — the square-mode residues, no others — while modulo 8 every odd square is 1 and those residues would be stranded. This is the same poverty of squares modulo powers of two that confines $D(2, \cdot)$ and $D(4, \cdot)$, one refinement up and just barely wealth enough. So "why $\{3, 8, 9\}$" has its complete answer — three moves, three jobs, zero waste: one governor $\equiv 8 \pmod{16}$ whose doubling is the modulus, plus a pair of odd rescuers whose residues cover all four standard-odd classes. And the number 9's double life upgrades rather than dissolves — rescuer in both modes, carrying $\{11, 15\}$ as a move and $\{9, 13\}$ as a square, the entire upper odd world leaning on the value 9.

The mechanism classifies tampering as well: an enlargement of the standard set leaves the P-set fixed exactly when every added move is either odd, landing, from any candidate position, on the all-N odd world, or $\equiv 8 \pmod{16}$, a clone of the governor. Verified: 1, 5, and 24 preserve the law, 2, 4, and 16 break it at $n = 2, 4, 16$, and the maximal enlargement, every odd number plus the whole clone class, holds to 2,000. The extension of Proposition 11.6 adds precisely the residue classes of 3, 9, and 8 themselves, squarely inside the harmless family, so the conjecture is not merely true but forced. It is busy rather than idle: below 3,000 the new moves serve as genuine witnesses at 1,490 positions, 746 of them odd, the smallest being $19 \to 0$, a new odd move rescuing its own residue. None of this replaces the verification proof, which remains the proof of record; it explains why the object that proof verified could not have looked any other way.

Proposition 11.11 (Odd fallbacks collapse)**.** *Let a be any odd positive integer and consider the mode-switching game in which squares are legal from $n \equiv 1 \pmod 2$ and the single fallback a is legal from even n. Then the P-positions are exactly the even integers, and the Grundy sequence is the same for every odd a — the fallback evaporates.*

Proof. Setup. Write $G(n)$ for the Grundy value. An even position has at most one option, $n - a$, which is odd since a is odd and is legal exactly when $n \geq a$. An odd position has as options the squares $n - k^2$ with $k^2 \leq n$; it always has at least the option $n - 1$, since odd $n \geq 1$, and $n - 1$ is even.

Part (i): every even n has $G(n) = 0$. Induct on n. For even n the reachable set of Grundy values is empty when $n < a$, and otherwise the single value $\{G(n-a)\}$ with $n-a$ odd and smaller. By Part (ii), applied at the smaller odd position $n-a$, that value is nonzero. The mex of the empty set is 0, and the mex of a one-element set of nonzero values is likewise 0, so $G(n) = 0$ in both cases.

Part (ii): every odd n has $G(n) \neq 0$. Let n be odd. Its option $n-1$ is even and smaller, so $G(n-1) = 0$ by Part (i), and a position with an option of Grundy value 0 has mex at least 1. Hence $G(n) \neq 0$.

The two parts induct together on n, each invoking the other only at strictly smaller positions, so the joint induction is legitimate and the P-positions — the positions of Grundy value 0 — are exactly the even integers.

Independence of a. Part (ii) never mentions a, and Part (i) uses a only to determine whether the even position has an option at all; either way the mex is 0. More strongly, the full Grundy value of an odd n is the mex of $\{G(n-k^2) : k^2 \leq n\}$, and by Parts (i) and (ii) each such value depends only on the parity of $n-k^2$ and on the Grundy values of smaller odd positions — never on a. By induction the entire Grundy sequence is therefore identical for every odd fallback a: the parameter is invisible to the game, because the squares already supply, from every odd position, a move to an even one, and the fallback can only ever move between residues that Part (i) has already flattened. $\square$

12 Discovery methodology and retrodictions

The conjectures in this paper were proposed by GameHunter, a closed loop of four parts: a proposer that varies rulesets, a Grundy engine that computes exact values, deterministic detectors that flag periodicity and modular structure, and a novelty filter that discards sequences already catalogued in the OEIS. Seeded evolutionary hunts surfaced the earliest solved games, Foursquare among them; systematic sweeps then enumerated the diagonal family, 1,120 games in the first sweep and 1,848 by the last, with exact laws read in 842 of them. Every law survived six adversarial verification passes at depths 20,000 and 10^6 without a failure, and a deeper rescan of the 731 resistant games converted only 6. Before and after every session the engine re-derives five classical laws (the subtraction game $\{1, 2, 3\}$ and Golomb's subtract-a-square among them), so that drift anywhere in the stack would announce itself in the classics first. The instrument stands in a short lineage of proposing machines — Graffiti [10], the Ramanujan Machine [11], FunSearch [12] — and inherits their division of labor rather than their architectures.

The division of labor between the instrument and the mathematics is strict, and the repository states it in writing: the instrument proposes and verifies mechanically and constructs no proofs, and each numerical claim above regenerates from the seeded scripts at `https://github.com/hamzagul07/gamehunter`, where the integrity protocol and its full session logs also live. How the proofs themselves were drafted is a separate question, and it is answered in the acknowledgments. The protocol earns its keep rather than asserting itself: one drafting error (a subtly different game verified faithfully across three documents) was caught only because the discipline requires sentences to follow printouts, never precede them.

Twice the instrument recorded numbers that no theory then existed to explain, and both entries were later derived exactly. The unexplained P-positions $3, 11, 23$ in $D(2, 1)$ and $5, 13, 37$ in $D(4, 1)$ are the images of Golomb's opening losing positions under the reindexings of Theorem 7.3, and the recorded preperiod 14 for $D(9, 4)$ is one more than the rebel position 13 of

Corollary 5.2. Both were readings before they were consequences, which is the strongest form of confirmation a proposing machine can offer: the prediction was on the record before the mechanism that implies it existed.

The traffic also ran the other way. Corollary 8.3 (that r enters the class-index Grundy sequence only through β) was derived before it was swept, and a sweep across eight moduli then returned the collapse: at each modulus, every healthy game with $r \geq 1$ produced one and the same class-index sequence, and the three exceptional games at $m = 9$ produced theirs. The Grundy Transfer law was confirmed against 68 games and 1.7 million positions, and the sweep's ledger is itself a receipt: its 77 disagreements are precisely the members of the excluded prefixes $\mathrm{Bad}(m, r)$ — the mismatches map the theorem's boundary rather than breaching it. And once, theory corrected a reading rather than confirming it: the embedding sweeps behind Section 9 reported hard-side exactness for every index, which was true of an index based at the second member and false at the first — the base-point clause of Theorem 9.5 exists because the proof caught what a base-blind sweep could not.

The closest antecedent we know is the work of Holshouser and Reiter on subtraction games whose move sets vary with the position [5], which anticipates the position-dependent format studied here while leaving the mode-switching family and its classification open.

13 What is closed, and what was always Golomb's

The question this section used to pose is answered inside the paper. The two columns the census could never fit — the confined classes of $D(2, r)$ and $D(4, r)$ — are Golomb's subtract-a-square game up to an affine change of coordinates (Theorem 7.3); they never settle into any eventual period (Corollary 7.5); and their P-positions thin toward density zero while never falling below the $\sqrt{n}/2$ floor (Corollary 7.6). The instrument's old verdict — lawless to 200,000, no modulus up to 200 — is upgraded from a reading to three theorems. The verdict was also, we can now say, slightly modest: the instrument's own density figures were carrying the answer all along. $D(2, 1)$ reads 0.2586 at 200,000 against the limiting 1/4 of Corollary 7.7, and the excess, 0.0086, is precisely the confined class's contribution — alive, thinning on schedule, and held above its $1/(2\sqrt{n})$ speed limit by the floor.

What remains open no longer belongs to this paper; it belongs to Golomb's game, where it has lived since 1966. The true growth rate of Golomb's P-set is wedged between two proven bounds — the $\sqrt{n}$ floor (Corollary 7.6, second half) and the $o(n)$ ceiling that Furstenberg–Sárközy places on any square-difference-free set (Corollary 7.6, first half). And our instrument reads a count near $n^{0.7}$ across its whole range: a reading, not a theorem, and one that must stay instrument-tier forever unless a genuinely new bound arrives, since no finite prefix can certify an exponent. The Grundy values of the same game (OEIS A014586) are more mysterious still [6, 7]. Problem A1 of the Games of No Chance list [8] asks how the structure of a ruleset constrains the structure of its periods; the diagonal family now answers on both sides of its own boundary. Where the palette exists, the law is complete and threshold-free, and Corollary 8.5 shows that even there the constraint is an outcome-level phenomenon only: every solved member of the family escapes the classical periodicity theory at the level of Grundy values. And the boundary itself is not a limitation of method: by Corollary 9.7, a classification theorem for mode-switching subtraction games with arbitrary mode sets would classify all subtraction games, so no such theorem exists to be proved. One construction away from the solved family lies a class as wild as subtraction games themselves.

We still leave the reader $D(2,1)$ as a parting gift. Subtract any square from an odd number; subtract 2 from an even one. A pencil reaches 3, 11, and 23 inside an afternoon — and this paper can now tell you what those numbers do forever: they never fall into a rhythm, and they grow rarer without ever running out. What it cannot tell you is the pace of the thinning. That question is older than this paper, and it is still open.

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References

[1] E. Duchêne and M. Rigo, Invariant games, *Theoret. Comput. Sci.* 411 (2010) 3169–3180.

[2] S. W. Golomb, A mathematical investigation of games of "take-away", *J. Combin. Theory* 1 (1966), no. 4, 443–458.

[3] U. Larsson and I. Saha, with an appendix by K. Suetsugu, A brief conversation about subtraction games, in *Games of No Chance 6*, Math. Sci. Res. Inst. Publ. 71, edited by U. Larsson, Cambridge Univ. Press, 2025, pp. 25–41.

[4] A. A. Siegel, Finite excluded subtraction sets and infinite modular Nim, M.Sc. thesis, Dalhousie University, 2005.

[5] A. Holshouser and H. Reiter, One pile Nim with arbitrary move function, *Electron. J. Combin.* 10 (2003), #N7.

[6] N. Fox, On aperiodic subtraction games with bounded Nim sequence, arXiv:1407.2823 (2014).

[7] U. Larsson and N. Fox, An aperiodic subtraction game of Nim-dimension two, *J. Integer Seq.* 18 (2015), Article 15.7.4.

[8] R. J. Nowakowski, Unsolved problems in combinatorial games, in *Games of No Chance 6* (U. Larsson, ed.), Math. Sci. Res. Inst. Publ. 71, Cambridge University Press, 2025, p. 43.

[9] M. H. Albert, R. J. Nowakowski, and D. Wolfe, *Lessons in Play: An Introduction to Combinatorial Game Theory*, 2nd ed., A K Peters/CRC Press, 2019.

[10] S. Fajtlowicz, On conjectures of Graffiti, *Discrete Math.* 72 (1988) 113–118.

[11] G. Raayoni et al., Generating conjectures on fundamental constants with the Ramanujan Machine, *Nature* 590 (2021) 67–73.

[12] B. Romera-Paredes et al., Mathematical discoveries from program search with large language models, *Nature* 625 (2024) 468–475.

[13] R. B. Austin, Impartial and partisan games, M.Sc. thesis, University of Calgary, 1976.

[14] N. B. Ho, On the expansion of three-element subtraction sets, *Theoret. Comput. Sci.* 582 (2015), 35–47.

[15] H. Furstenberg, Ergodic behavior of diagonal measures and a theorem of Szemerédi on arithmetic progressions, *J. Analyse Math.* 31 (1977), 204–256.

[16] A. Sárközy, On difference sets of sequences of integers. I, *Acta Math. Acad. Sci. Hungar.* 31 (1978), 125–149.

[17] I. Z. Ruzsa, Difference sets without squares, *Period. Math. Hungar.* 15 (1984), 205–209.

[18] D. Eppstein, Faster evaluation of subtraction games, in *FUN 2018*, LIPIcs 100, Art. 20, pp. 20:1–20:12.

[19] T. E. Plambeck and A. N. Siegel, Misère quotients for impartial games, *J. Combin. Theory Ser. A* 115 (2008), 593–622.

[20] U. Larsson, The $\star$-operator and invariant subtraction games, *Theoret. Comput. Sci.* 422 (2012), 52–58.

[21] U. Larsson, P. Hegarty and A. S. Fraenkel, Invariant and dual subtraction games resolving the Duchêne–Rigo conjecture, *Theoret. Comput. Sci.* 412 (2011), 729–735.